

Why Parallel Processing

– Grand Challenge

- **Quantum chemistry**
- **Cosmology and astrophysics**
- **Computational fluid dynamic and turbulence**
- **Materials design and superconductivity**
- **Biology, pharmacology, etc**
- **Medicine and modeling of human organs and bones**
- **Global weather and environmental modeling**

Evolution of Modern Computers

Table 1.1 Five Generations of Electronic Computers

Generation	Technology and Architecture	Software and Applications	Representative Systems
First (1945–54)	Vacuum tubes and relay memories, CPU driven by PC and accumulator, fixed-point arithmetic.	Machine/assembly languages, single user, no subroutine linkage, programmed I/O using CPU.	ENIAC, Princeton IAS, IBM 701.
Second (1955–64)	Discrete transistors and core memories, floating-point arithmetic, I/O processors, multiplexed memory access.	HLL used with compilers, subroutine libraries, batch processing monitor.	IBM 7090, CDC 1604, Univac LARC.
Third (1965–74)	Integrated circuits (SSI/-MSI), microprogramming, pipelining, cache, and lookahead processors.	Multiprogramming and time-sharing OS, multiuser applications.	IBM 360/370, CDC 6600, TI-ASC, PDP-8.
Fourth (1975–90)	LSI/VLSI and semiconductor memory, multiprocessors, vector supercomputers, multicomputers.	Multiprocessor OS, languages, compilers, and environments for parallel processing.	VAX 9000, Cray X-MP, IBM 3090, BBN TC2000.
Fifth (1991–present)	ULSI/VHSIC processors, memory, and switches, high-density packaging, scalable architectures.	Massively parallel processing, grand challenge applications, heterogeneous processing.	Fujitsu VPP500, Cray/MPP, TMC/CM-5, Intel Paragon.

Elements of a Modern Computer System

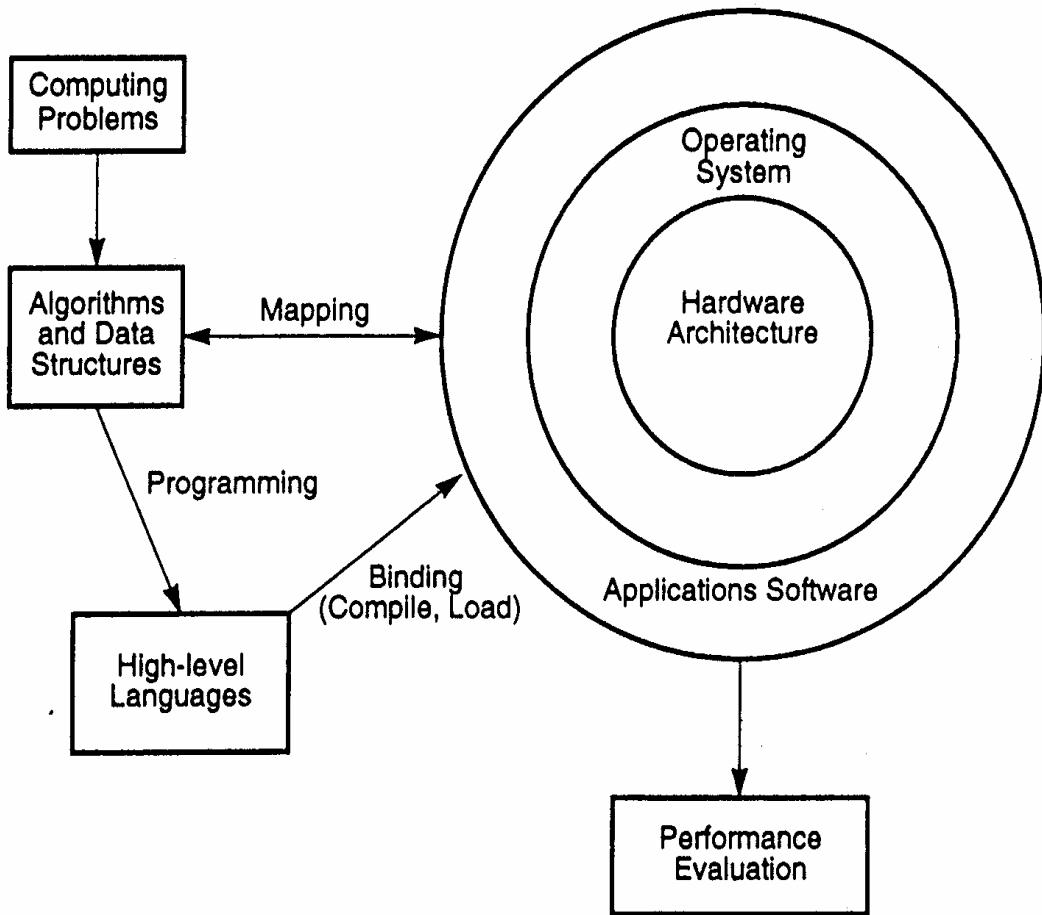
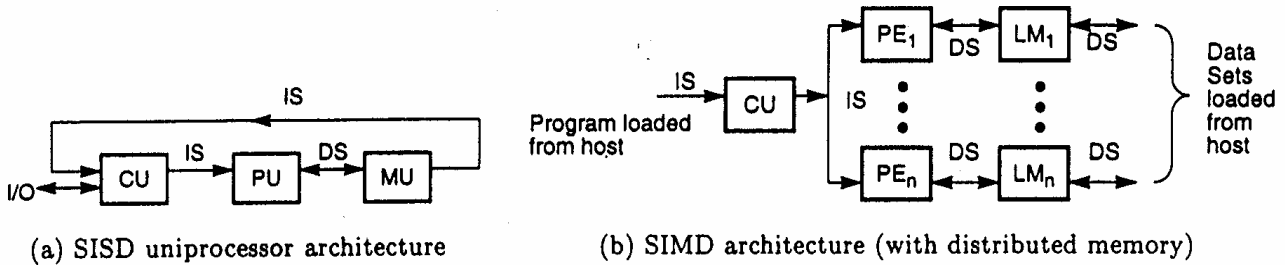


Figure 1.1 Elements of a modern computer system.

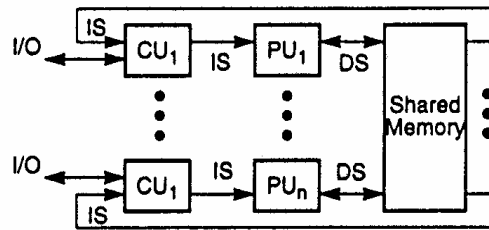
Classification of Computer Architecture

– Flynn's Classification

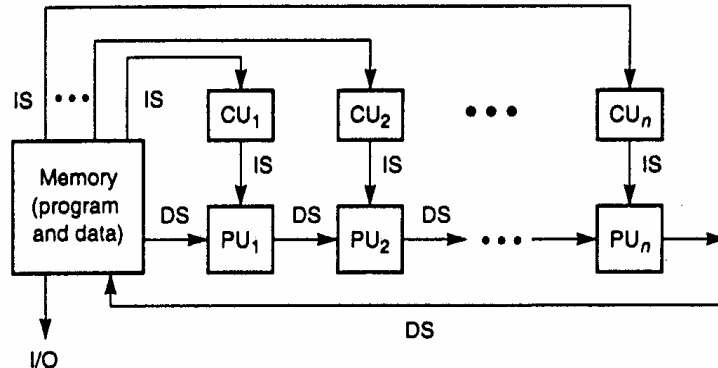


Captions:

CU = Control Unit
 PU = Processing Unit
 MU = Memory Unit
 IS = Instruction Stream
 DS = Data Stream
 PE = Processing Element
 LM = Local Memory



(c) MIMD architecture (with shared memory)



(d) MISD architecture (the systolic array)

Figure 1.3 Flynn's classification of computer architectures. (Derived from Michael Flynn, 1972)

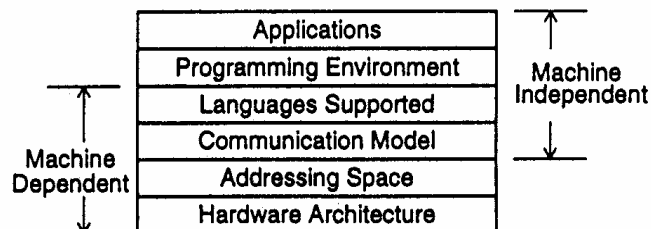


Figure 1.4 Six layers for computer system development. (Courtesy of Lionel Ni, 1990)

Evolution of Computer Architecture

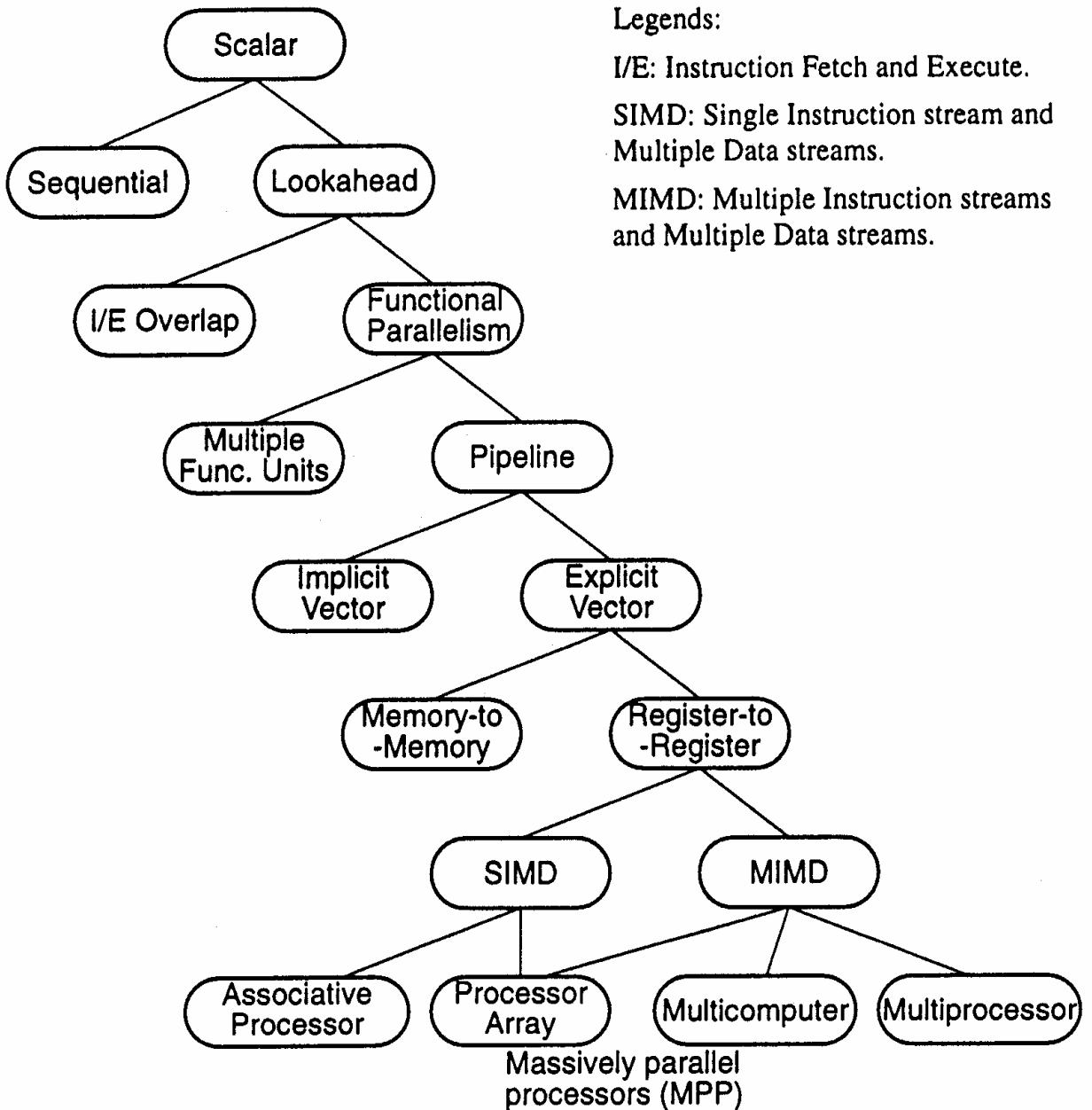


Figure 1.2 Tree showing architectural evolution from sequential scalar compute to vector processors and parallel computers.

Shared-Memory Architecture

– The UMA Model

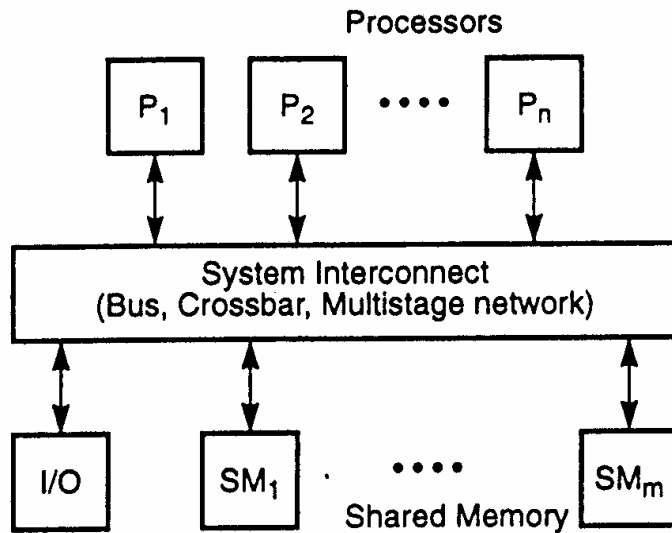
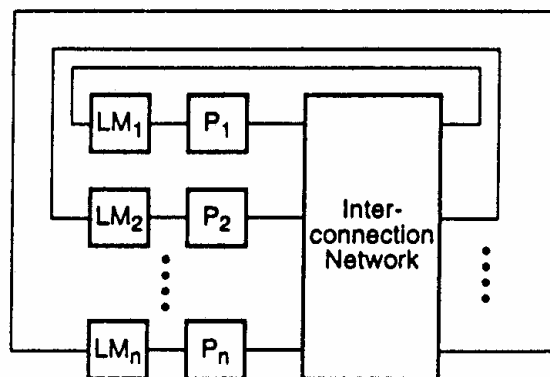


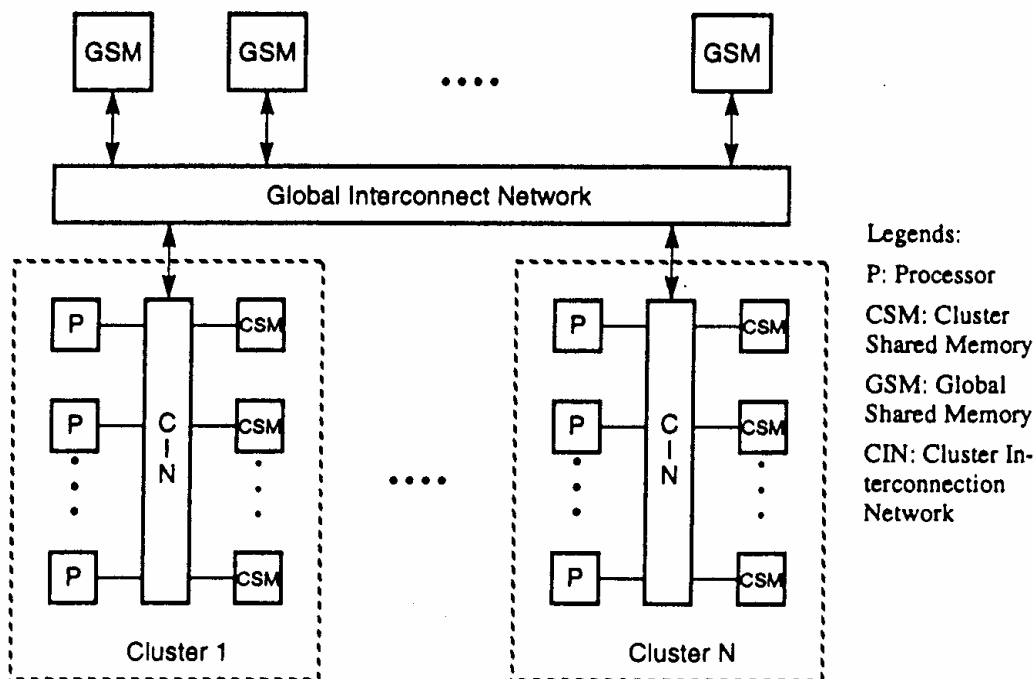
Figure 1.6 The UMA multiprocessor model (e.g., the Sequent Symmetry S-81).

Shared-Memory Architecture (Cont'd)

– The NUMA Model



(a) Shared local memories (e.g., the BBN Butterfly)



(b) A hierarchical cluster model (e.g., the Cedar system at the University of Illinois)

Figure 1.7 Two NUMA models for multiprocessor systems.

Shared-Memory Architecture (Cont'd)

- The COMA Model

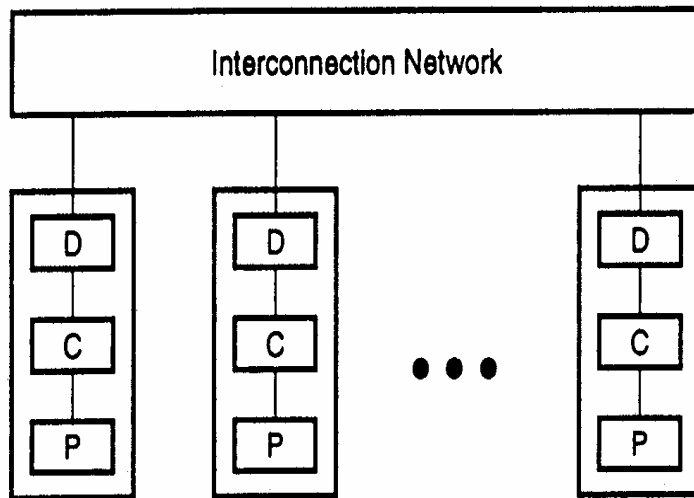


Figure 1.8 The COMA model of a multiprocessor. (P: Processor, C: Cache, D: Directory; e.g., the KSR-1)

Distributed-Memory Architecture

– Generic Model

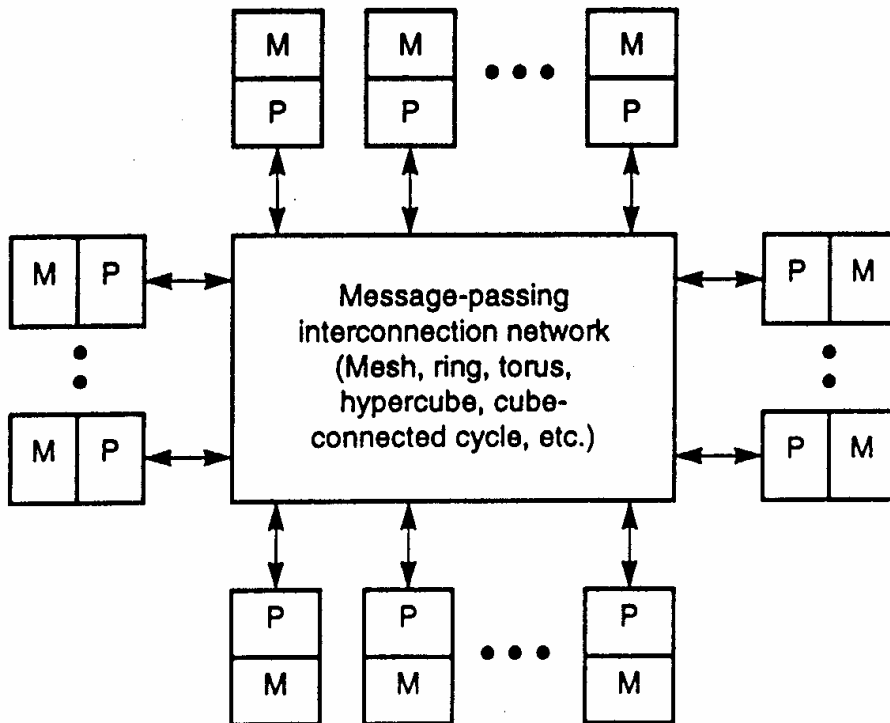
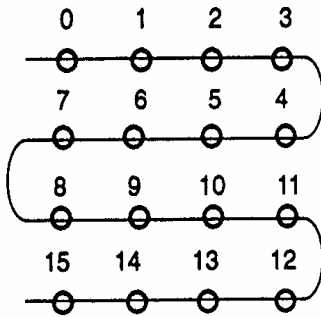


Figure 1.9 Generic model of a message-passing multicomputer.

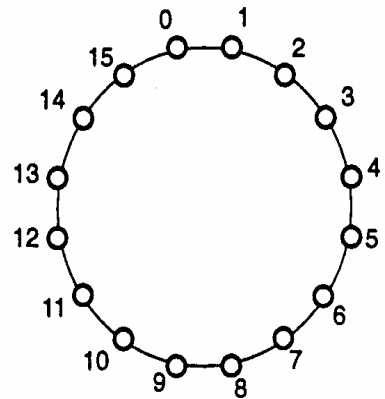
Distributed-Memory Architecture (Cont'd)

– Static Interconnection Networks

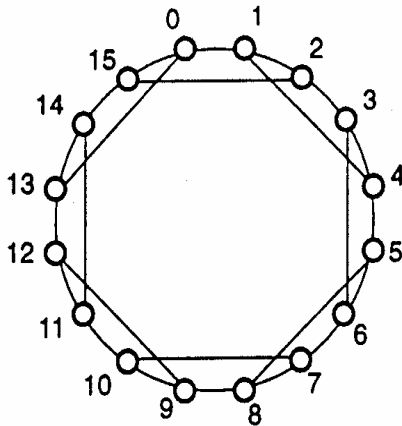
• Linear Array, Ring, Chordal Ring, Barrel Shifter



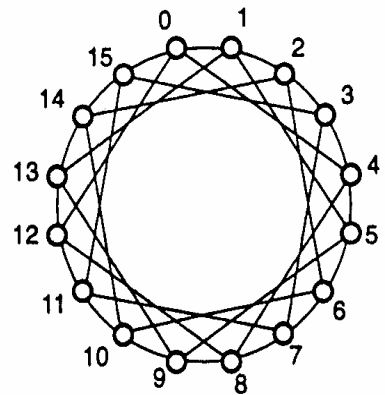
(a) Linear array



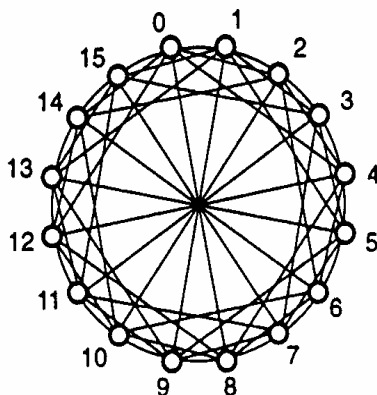
(b) Ring



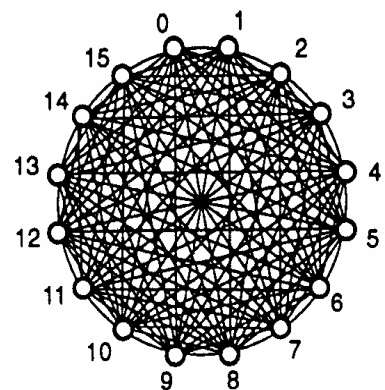
(c) Chordal ring of degree 3



(d) Chordal ring of degree 4 (same as Illiac mesh)



(e) Barrel shifter

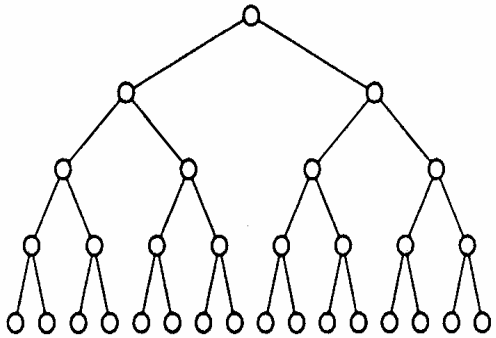


(f) Completely connected

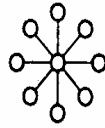
Distributed-Memory Architecture (Cont'd)

– Static Interconnection Networks

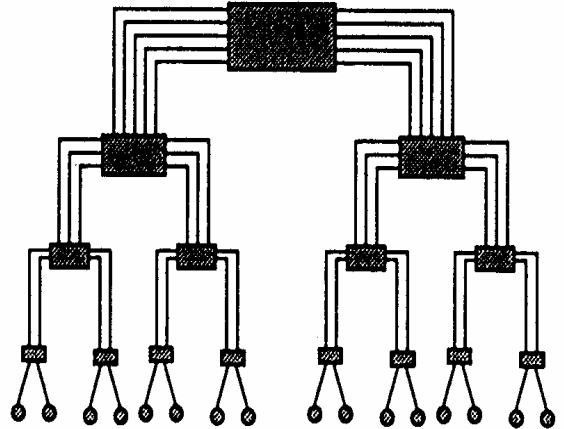
- Tree, Star, Fat Tree



(a) Binary tree



(b) Star



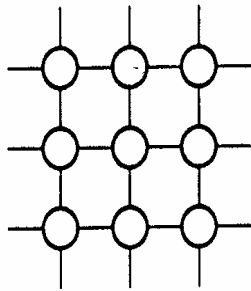
(c) Binary fat tree

Figure 2.17 Tree, star, and fat tree.

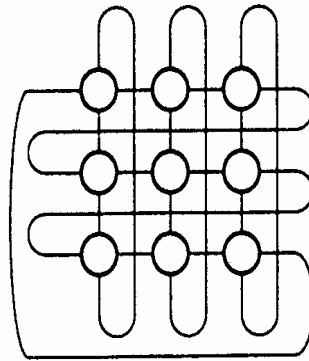
Distributed-Memory Architecture (Cont'd)

– Static Interconnection Networks

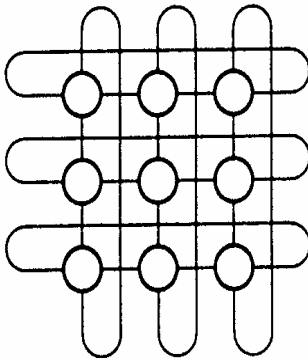
- Mesh, Torus, Systolic Array



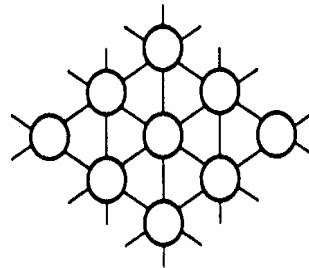
(a) Mesh



(b) Illiac mesh



(c) Torus



(d) Systolic array

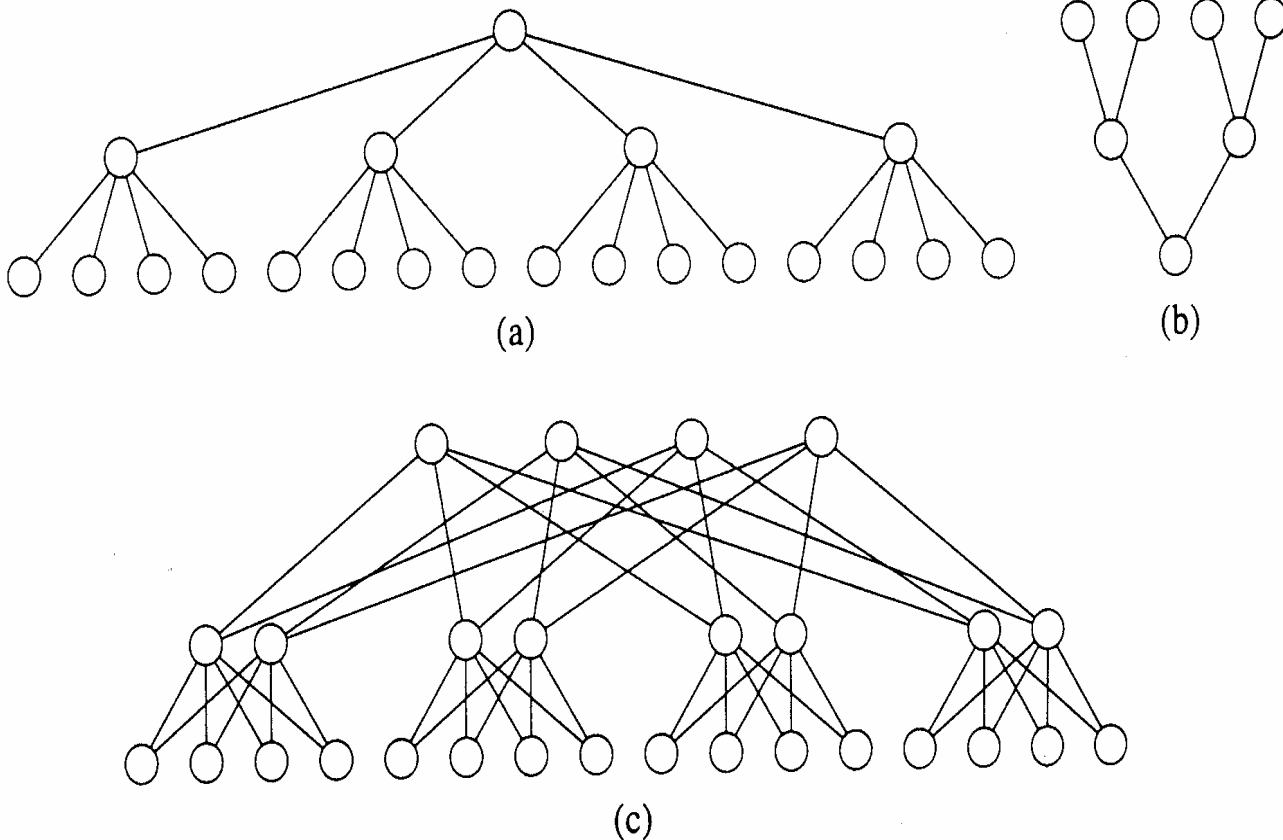
Figure 2.18 Mesh, Illiac mesh, torus, and systolic array.

Distributed-Memory Architecture (Cont'd)

– Static Interconnection Networks

• Hypertree Networks

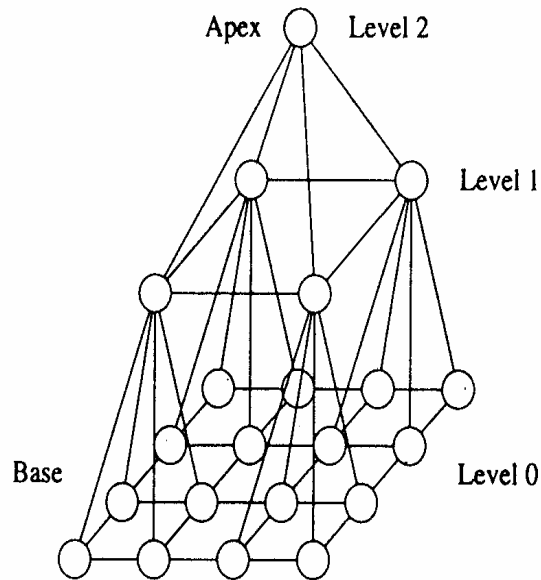
FIGURE 3-3 Hypertree network of degree 4 and depth 2. (a) Front view. (b) Side view. (c) Complete network.



Distributed-Memory Architecture (Cont'd)

– Static Interconnection Networks

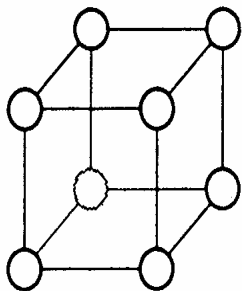
- Pyramid Networks



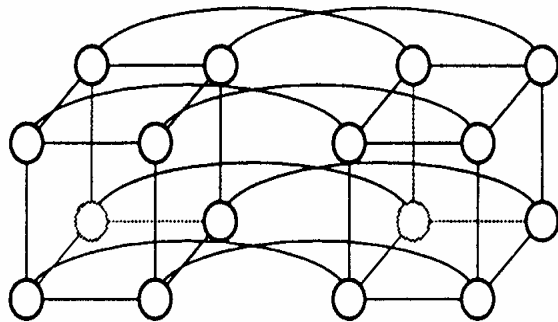
Distributed-Memory Architecture (Cont'd)

– Static Interconnection Networks

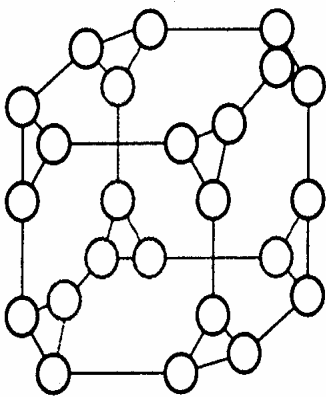
- Hypercube, Cube-Connected Cycles



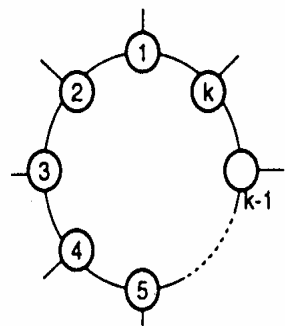
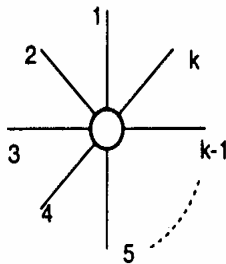
(a) 3-cube



(b) A 4-cube formed by interconnecting two 3-cubes



(c) 3-cube-connected cycles



(d) Replacing each node of a k -cube by a ring (cycle) of k nodes to form the k -cube-connected cycles

Figure 2.19 Hypercubes and cube-connected cycles.

Distributed-Memory Architecture (Cont'd)

– Dynamic Interconnection Networks

• Bus

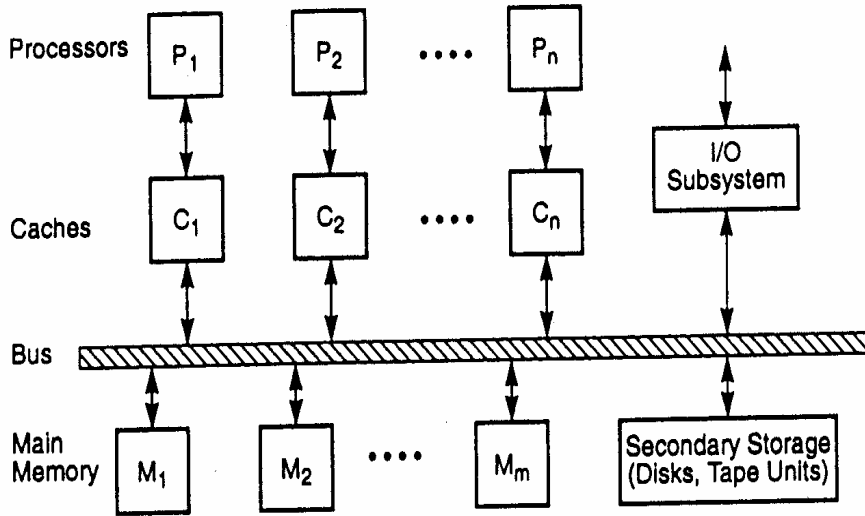


Figure 2.22 A bus-connected multiprocessor system, such as the Sequent Symmetry S1.

Distributed-Memory Architecture (Cont'd)

– Dynamic Interconnection Networks

• Generic MIN Network

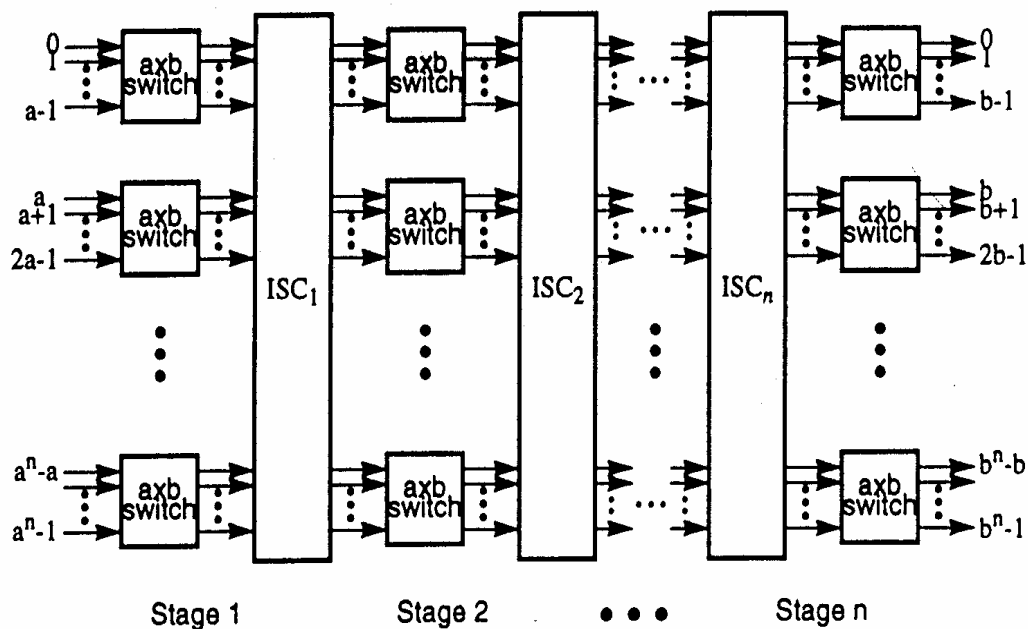


Figure 2.23 A generalized structure of a multistage interconnection network (MIN) built with $a \times b$ switch modules and interstage connection patterns $ISC_1, ISC_2, \dots, ISC_n$.

Distributed-Memory Architecture (Cont'd)

– Dynamic Interconnection Networks

• Omega Network

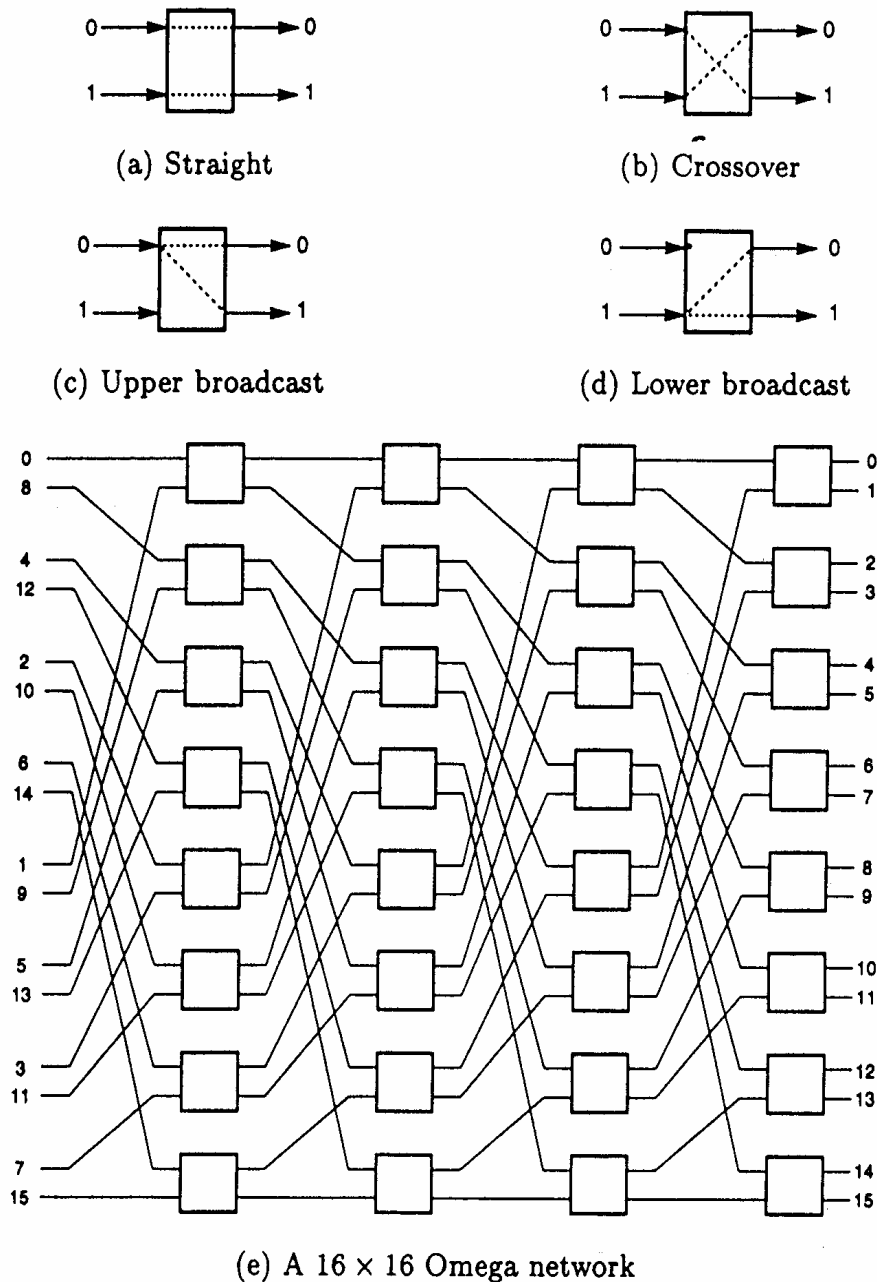
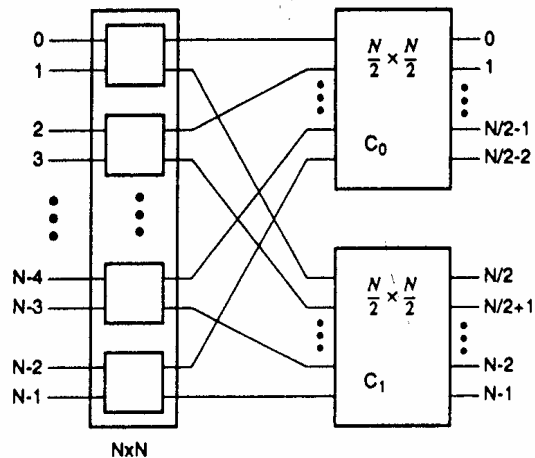


Figure 2.24 The use of 2x2 switches and perfect shuffle as an interstage connection pattern to construct a 16x16 Omega network. (Courtesy of Duncan Lawrie; reprinted with permission from *IEEE Trans. Computers*, Dec. 1975)

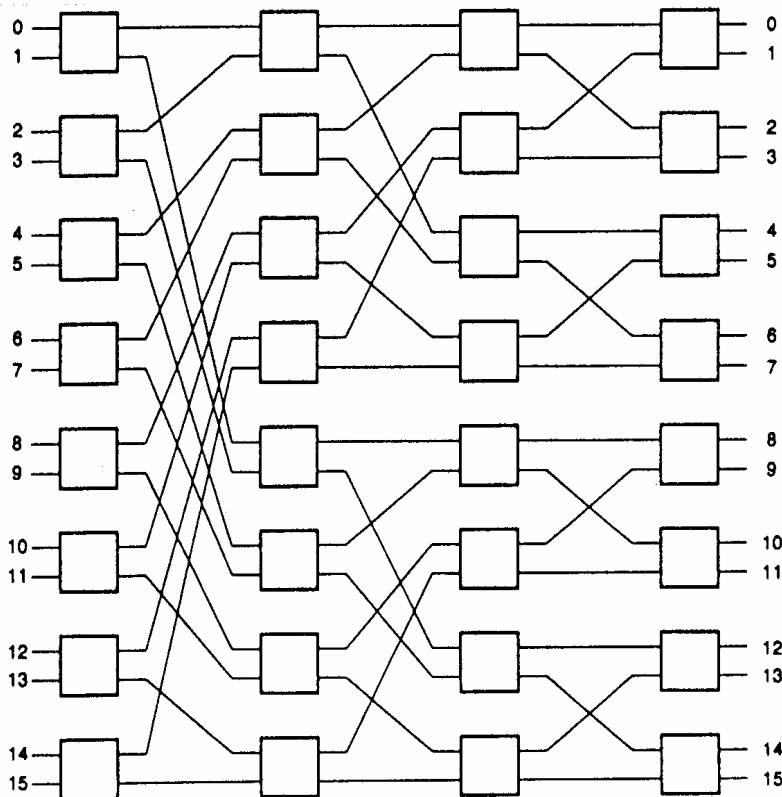
Distributed-Memory Architecture (Cont'd)

– Dynamic Interconnection Networks

• Baseline Network



(a) Recursive construction

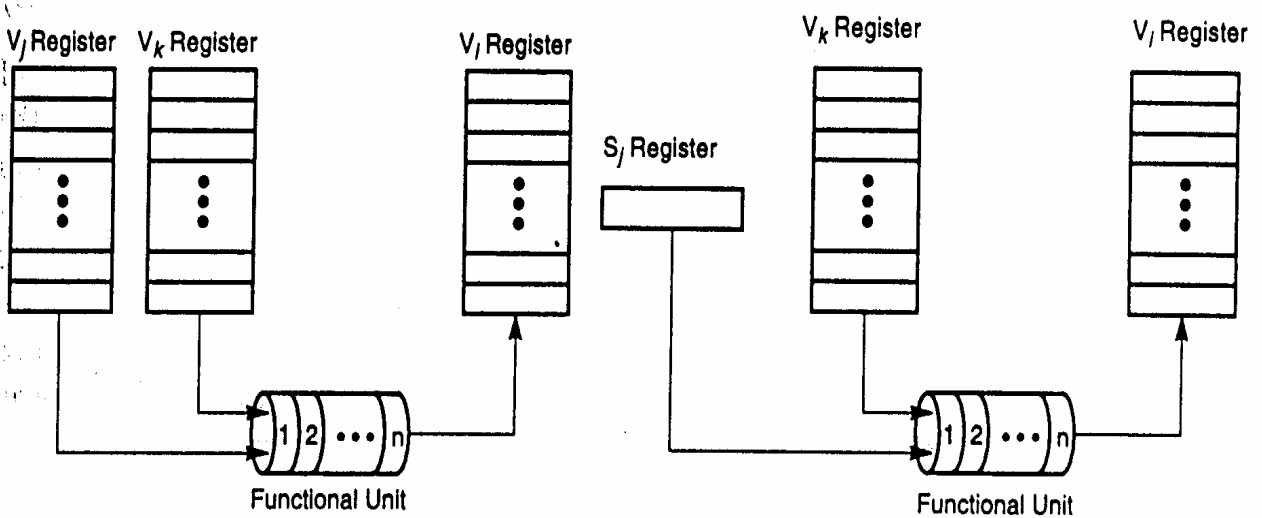


(b) A 16×16 Baseline network

Figure 2.25 Recursive construction of a Baseline network. (Courtesy of Wu and Feng; reprinted with permission from *IEEE Trans. Computers*, August 1980)

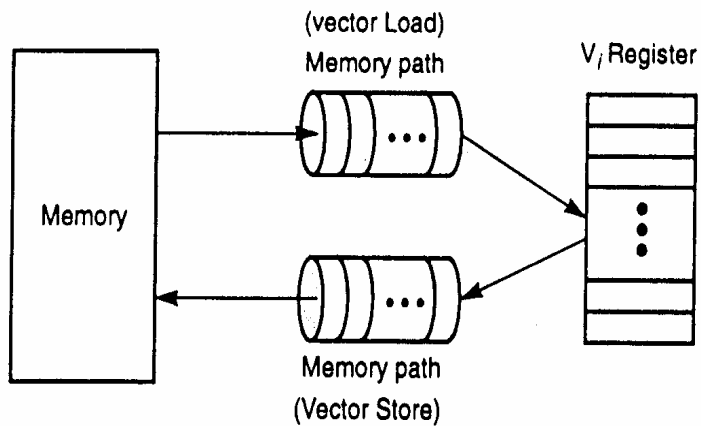
Vector Processors

– Vector Instruction Types



(a) Vector-vector instruction

(b) Vector-scalar instruction

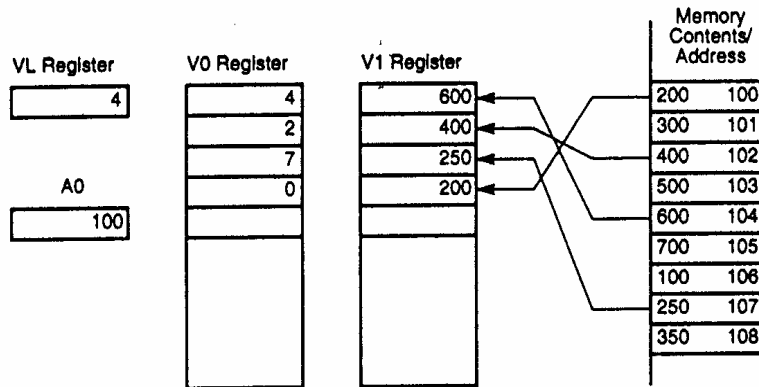


(c) Vector-memory instructions

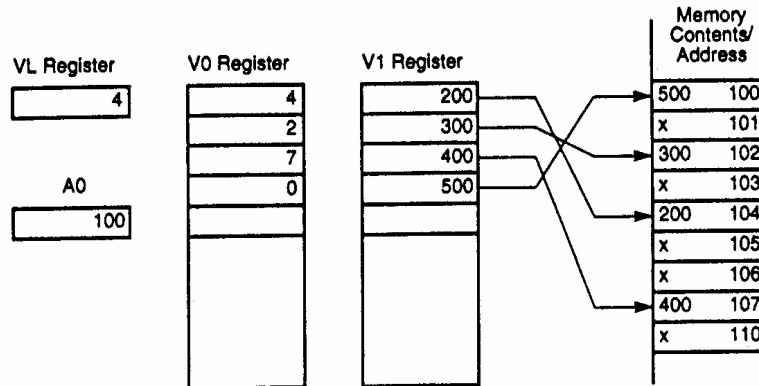
Figure 8.1 Vector instruction types in Cray-like computers.

Vector Processors (Cont'd)

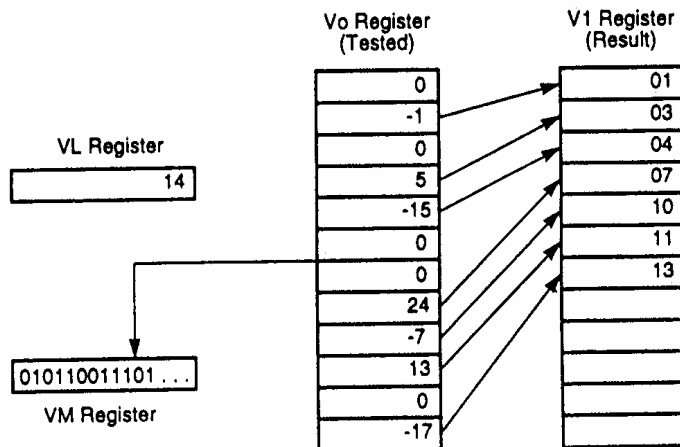
– Vector Instruction Types (Example)



(a) Gather instruction



(b) Scatter instruction



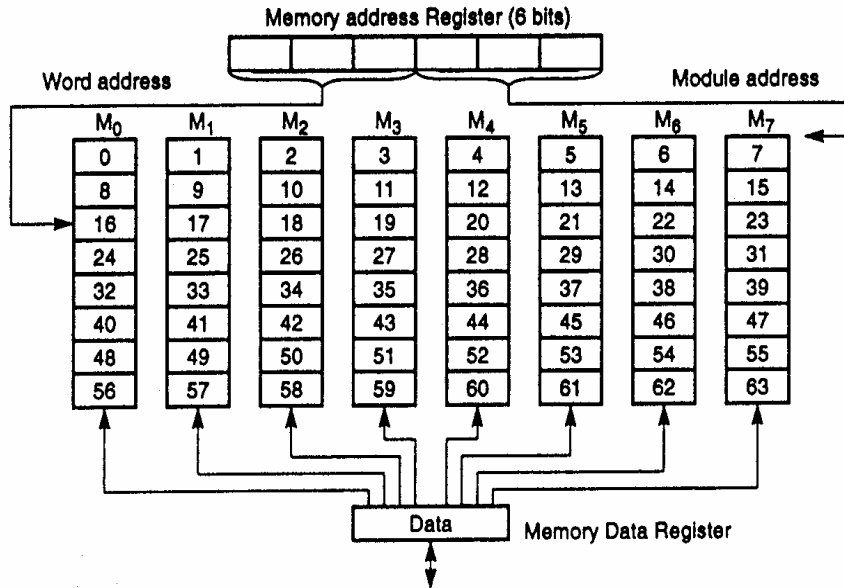
(c) Masking instruction

Figure 8.2 Gather, scatter and masking operations on the Cray Y-MP. (Courtesy of Cray Research Inc., 1990)

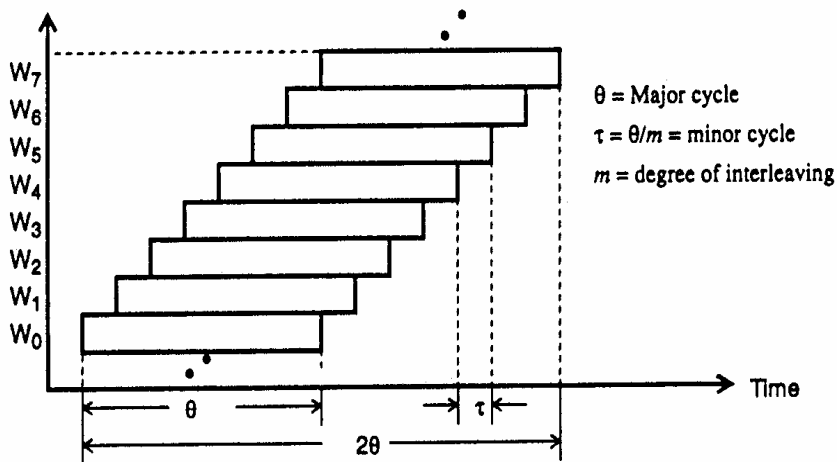
Vector Processors (Cont'd)

– Vector–Access Memory Schemes

• C–Access Memory Organization



(a) Eight-way low-order interleaving (absolute address shown in each memory word)



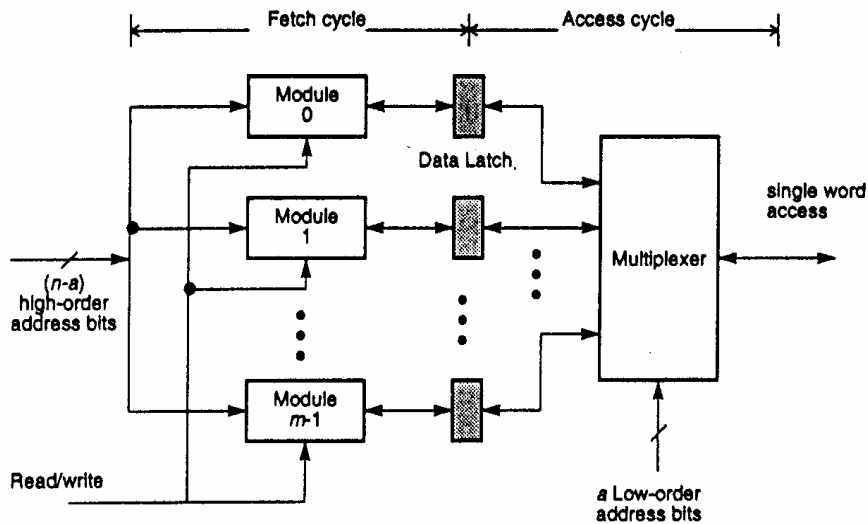
(b) Pipelined access of eight consecutive words in a C-access memory

Figure 5.16 Multiway interleaved memory organization and the C-access timing chart.

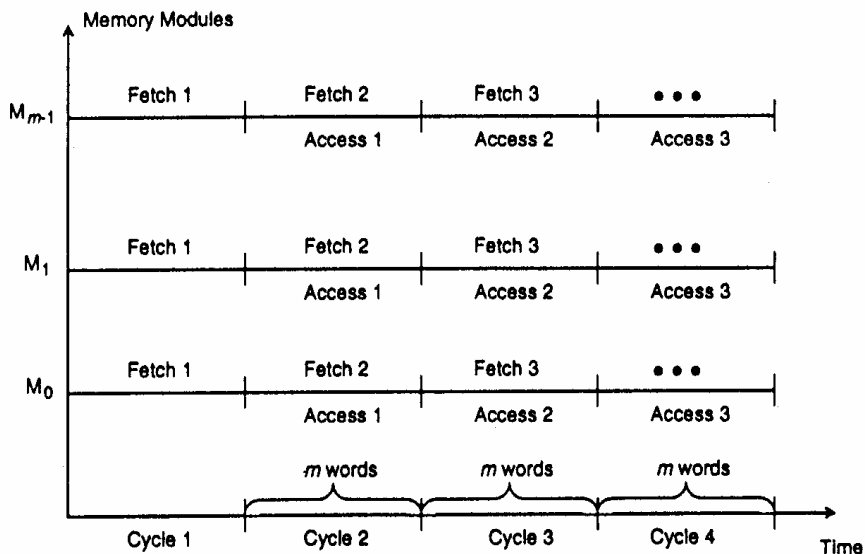
Vector Processors (Cont'd)

– Vector–Access Memory Schemes

• S–Access Memory Organization



(a) S-access organization for an m -way interleaved memory



(b) Successive vector accesses using overlapped fetch and access cycles

Figure 8.3 The S-access interleaved memory for vector operands access.

Vector Processors (Cont'd)

– Generic Architecture of Vector Supercomputer

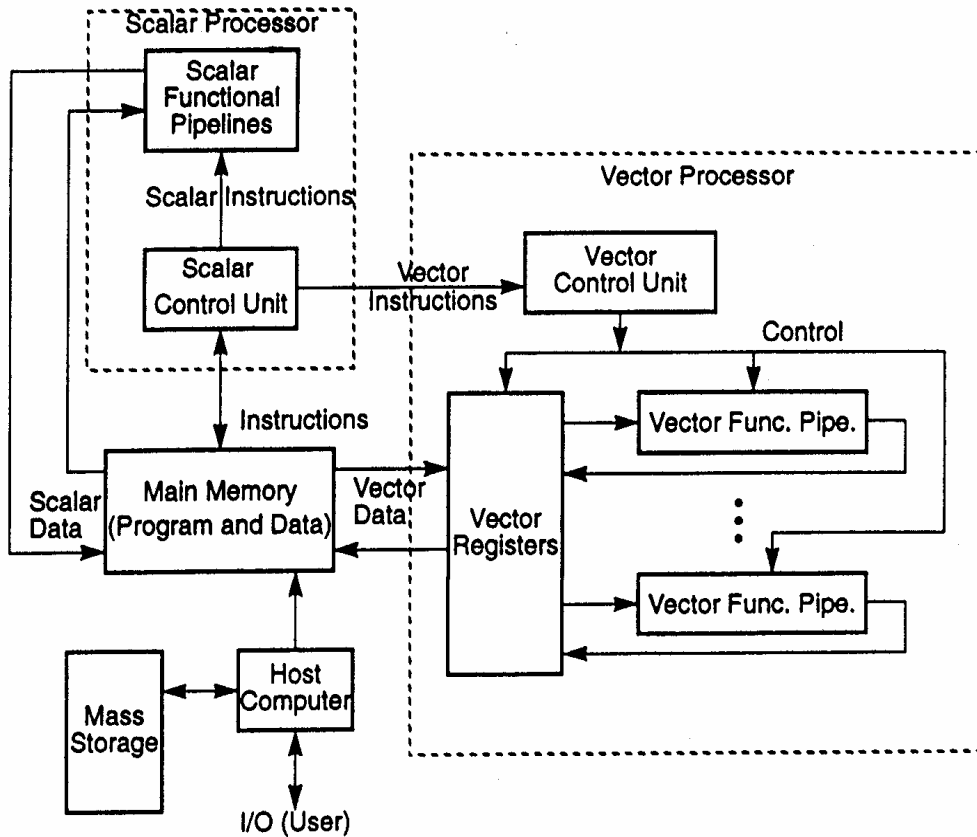


Figure 1.11 The architecture of a vector supercomputer.

Vector Processors (Cont'd)

– Vector–Access Memory Schemes

- C/S–Access Memory Organization

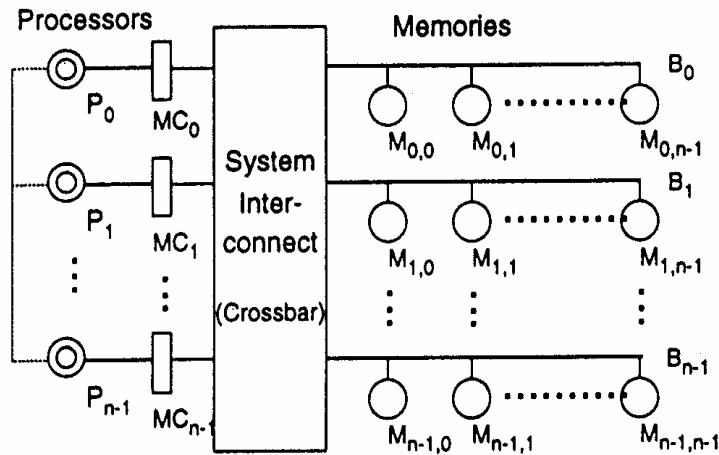


Figure 8.4 The C/S memory organization. (Courtesy of D.K. Panda, 1990)

Vector Processors (Cont'd)

– Past and Present Supercomputers

• Summary

Table 8.1 Summary of Full-Scale Supercomputers Made in the United States and in Japan

System model	Maximum configuration, clock rate, OS/Compiler	Unique features and remarks
Cray 1S	Uniprocessor with 10 pipelines, 12.5 ns, COS/CF77 2.1.	First ECL-based super, introduced in 1976.
Cray 2S /4-256	4 processors with 256M-word memory, 4.1 ns, COS or UNIX/CF77 3.0.	16K-word local memory, ported UNIX V introduced in 1985.
Cray X-MP 416	4 processors with 16M-word memory, and 128M-word SSD, 8.5 ns, COS CF77 5.0.	Using shared register clusters for IPC, introduced 1983.
Cray Y-MP 832	8 processors with 128M-word memory, 6 ns, CF77 5.0.	Enhanced from X-MP, introduced 1988.
Cray Y-MP C-90	16 processors with 2 vector pipes per processor, 4.2 ns, UNICOS/CF 77 5.0.	The largest Cray machine, introduced in 1991.
CDC Cyber 205	Uniprocessor with 4 pipelines, 20 ns, virtual OS/FTN 200.	Memory-to-memory architecture, introduced in 1982.
ETA 10 E	Uniprocessor with 10.5 ns, ETAV/FTN 200	Successor to Cyber 205, introduced in 1985.
NEC SX-X/44	4 processors with 4 sets of pipelines per processor, 2.9 ns, F77SX.	Succeeded by SX-X Series, introduced in 1991.
Fujitsu VP2600/10	Uniprocessor with 5 vector pipes and dual scalar processors, 3.2 ns, MSP-EX/F77 EX/VP.	Use reconfigurable vector registers and masking, introduced in 1991.
Hitachi 820/80	18 functional pipelines in a uniprocessor with 512 Mbytes memory 4 ns, FORT 77/HAP V23-OC.	Introduced in 1987 with 64 I/O channels providing a maximum of 288 Mbytes/s transfer.

Vector Processors (Cont'd)

– Past and Present Supercomputers

- Performance analysis

Table 8.2 Vector and Scalar Performance of Various Supercomputers

Machine	Cray 1S	Cray 2S	Cray X-MP	Cray Y-MP	Hitachi S820	NEC SX2	Fujitsu VP400
Vector performance (Mflops)	85.0	151.5	143.3	201.6	737.3	424.2	207.1
Scalar performance (Mflops)	9.8	11.2	13.1	17.0	17.8	9.5	6.6
Vector balance point	0.90	0.93	0.92	0.92	0.98	0.98	0.97

Source: J.E. Smith et al., Future General-Purpose Supercomputing Conference, *IEEE Supercomputing Conference*, 1990.

Vector Processors (Cont'd)

– Past and Present Supercomputers

• Cray Y-MP 816

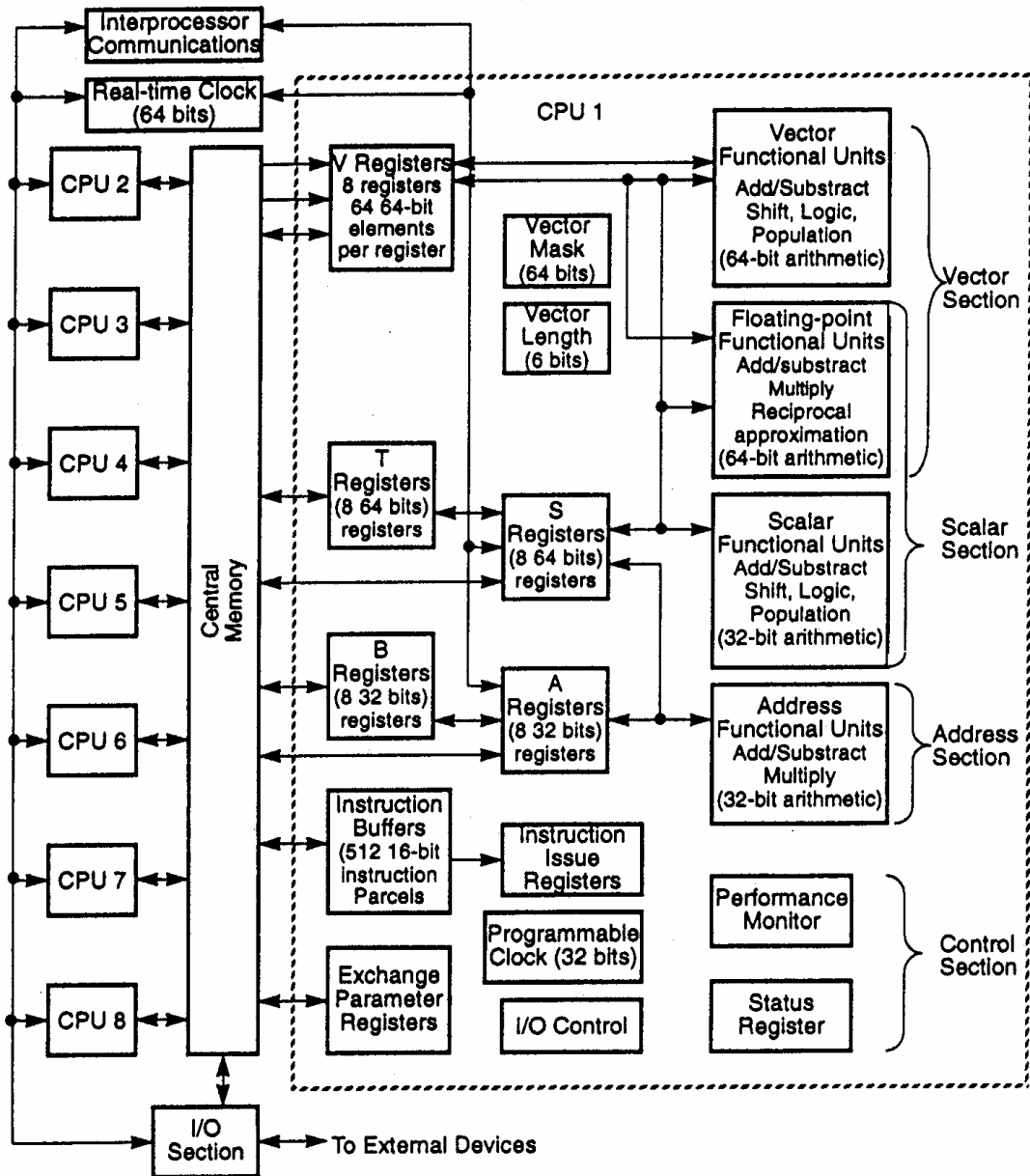


Figure 8.9 Cray Y-MP 816 system organization. (Courtesy of Cray Research, Inc., 1991)

Vector Processors (Cont'd)

– Past and Present Supercomputers

• Fujitsu VP2000

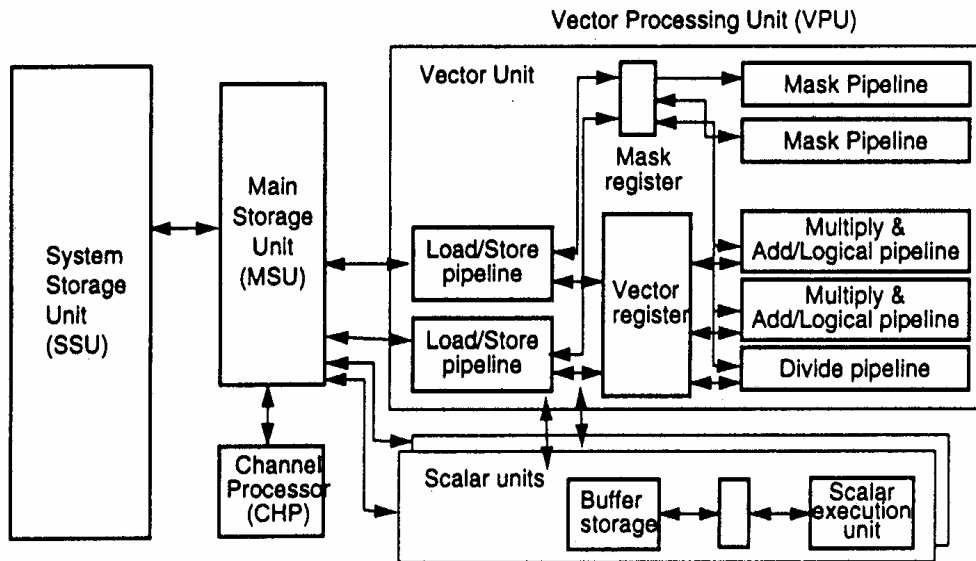


Figure 8.13 The Fujitsu VP2000 Series supercomputer architecture. (Courtesy of Fujitsu Ltd., 1991)

Vector Processors (Cont'd)

– Past and Present Supercomputers

• High-end Mainframe Supercomputers

Table 8.4 High-end Mainframe Supercomputers

Machine Characteristics	IBM ES/9000 –900 VF	DEC VAX 9000/440 VP	CDC Cyber 2000V
Number of processors	6 processors each attached to a vector facility	4 processors with vector boxes	2 central processors with vector hardware
Machine cycle time	9 ns	16 ns	9 ns
Maximum memory	1 Gbyte	512 Mbytes	512 Mbytes
Extended memory	8 Gbytes	2 Gbytes	N/A
Processor architecture: vector, scalar, and other functional units	Vector facility (VF) attached to each processor, delivering 4 floating-point results per cycle.	Vector processor (VBOX) connected to a scalar CPU. Two vector pipelines per VBOX. Four functional units in scalar CPU.	FPU for add and multiply, scalar unit with divide and multiply, integer unit and business data handler per processor.
I/O subsystem	256 ESCON fiber optic channels.	4 XMI I/O buses and 14 VAXBI I/O buses.	18 I/O processors with optional 18 additional I/O processors.
Operating system	MVS/ESA, VM/ESA, VSE/ESA	VMS or ULTRIX	NOS/VE
Vectorizing languages / compilers	Fortran V2 with interactive vectorization.	VAX Fortran compiler supporting concurrent scalar and vector processing.	Cyber 2000 Fortran V2.
Peak performance and remarks	2.4 Gflops (predicted).	500 Mflops peak.	210 Mflops per processor.

SIMD Computers

– Operational Model of SIMD Computers

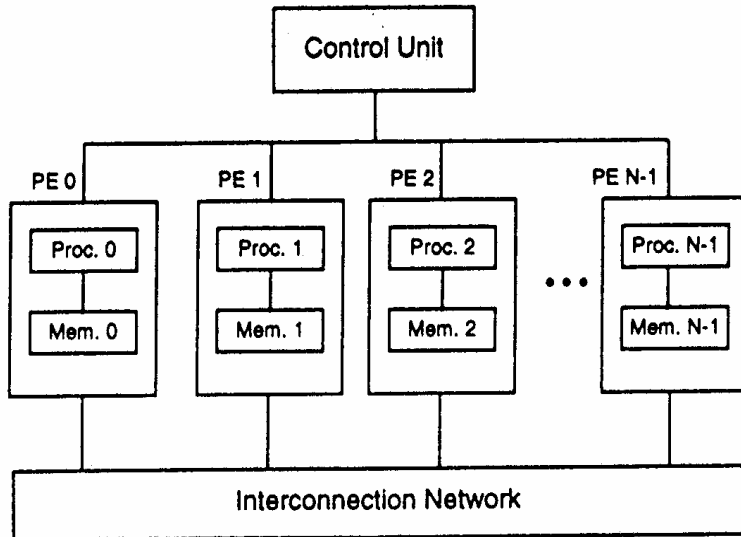
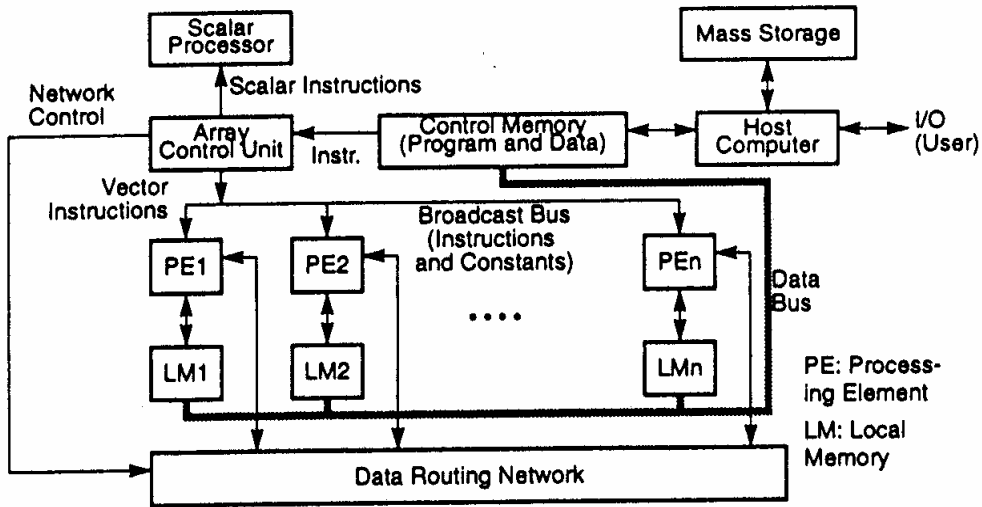


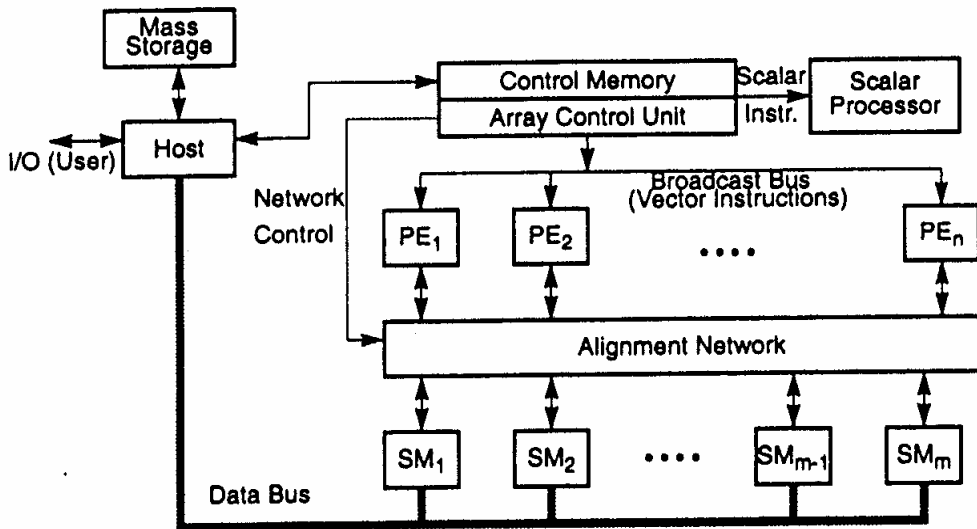
Figure 1.12 Operational model of SIMD computers.

SIMD Computers (Cont'd)

- SIMD Machine Models



(a) Using distributed local memories (e.g., the Illiac IV)



(b) Using shared-memory modules (e.g., the BSP)

Figure 8.22 Two models for constructing SIMD supercomputers.

SIMD Computers (Cont'd)

– Representative SIMD Supercomputers

Table 1.6 Representative SIMD Supercomputers

System Model	SIMD Machine Architecture and Capabilities	Languages, Compilers and Software Support
MasPar Computer Corporation MP-1 Family	Available in configurations from 1024 to 16,384 processors with 26,000 MIPS or 1.3 Gflops. Each PE is a RISC processor, with 16 Kbytes local memory. An X-Net mesh plus a multistage crossbar interconnect.	Fortran 77, MasPar Fortran (MPF), and MasPar Parallel Application Language; UNIX/OS with X-window, symbolic debugger, visualizers and animators.
Thinking Machines Corporation, CM-2	A bit-slice array of up to 65,536 PEs arranged as a 10-dimensional hypercube with 4×4 mesh on each vertex, up to 1M bits of memory per PE, with optional FPU shared between blocks of 32 PEs. 28 Gflops peak and 5.6 Gflops sustained.	Driven by a host of VAX, Sun, or Symbolics 3600, Lisp compiler, Fortran 90, C*, and *Lisp supported by PARIS
Active Memory Technology DAP600 Family	A fine-grain, bit-slice SIMD array of up to 4096 PEs interconnected by a square mesh with 1K bits per PE, orthogonal and 4-neighbor links, 20 GIPS and 560 Mflops peak performance.	Provided by host VAX/VMS or UNIX Fortran-plus or APAL on DAP, Fortran 77 or C on host. Fortran-plus affected Fortran 90 standards.

SIMD Computers (Cont'd)

– The CM-2 Architecture

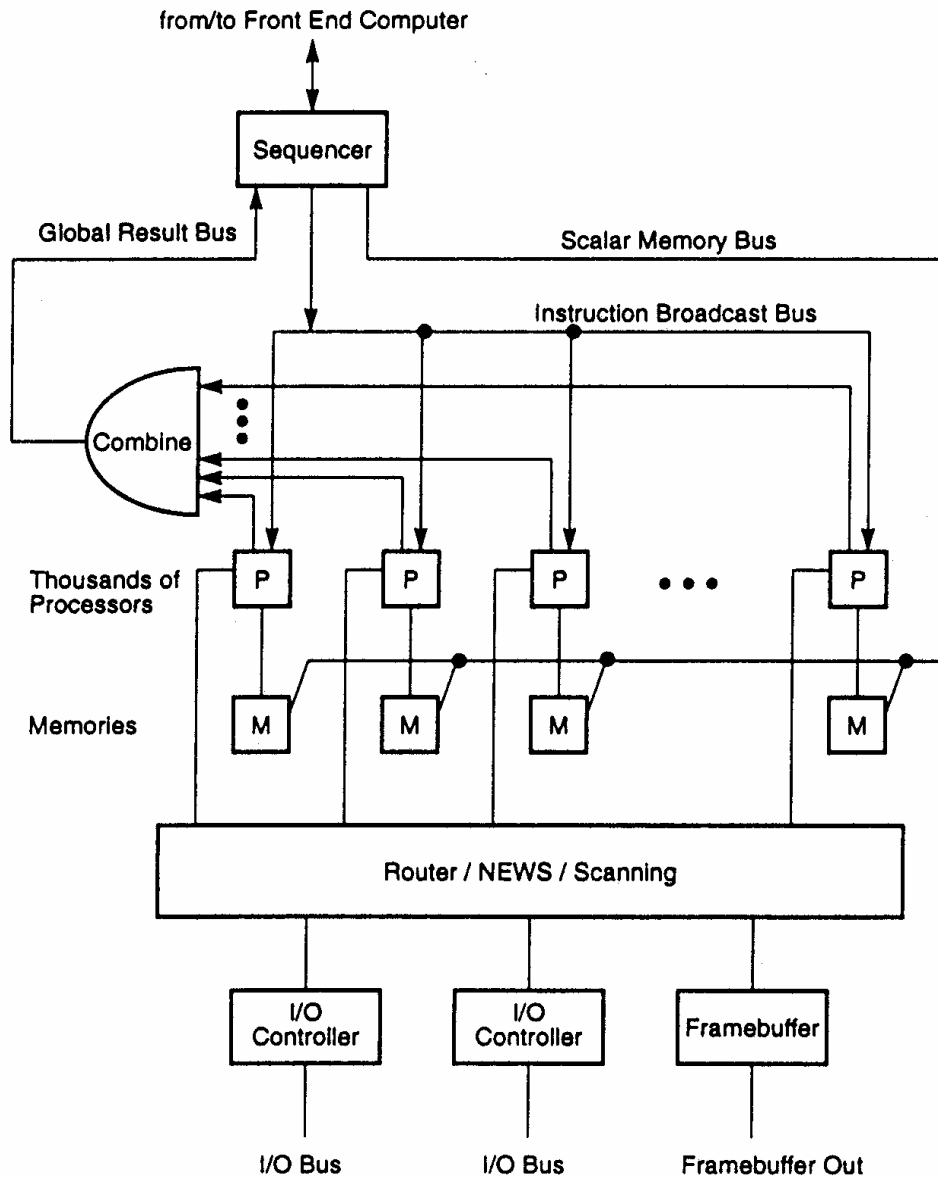


Figure 8.23 The architecture of the Connection Machine CM-2. (Courtesy of Thinking Machines Corporation, 1990)

SIMD Computers (Cont'd)

– The CM-2 Architecture

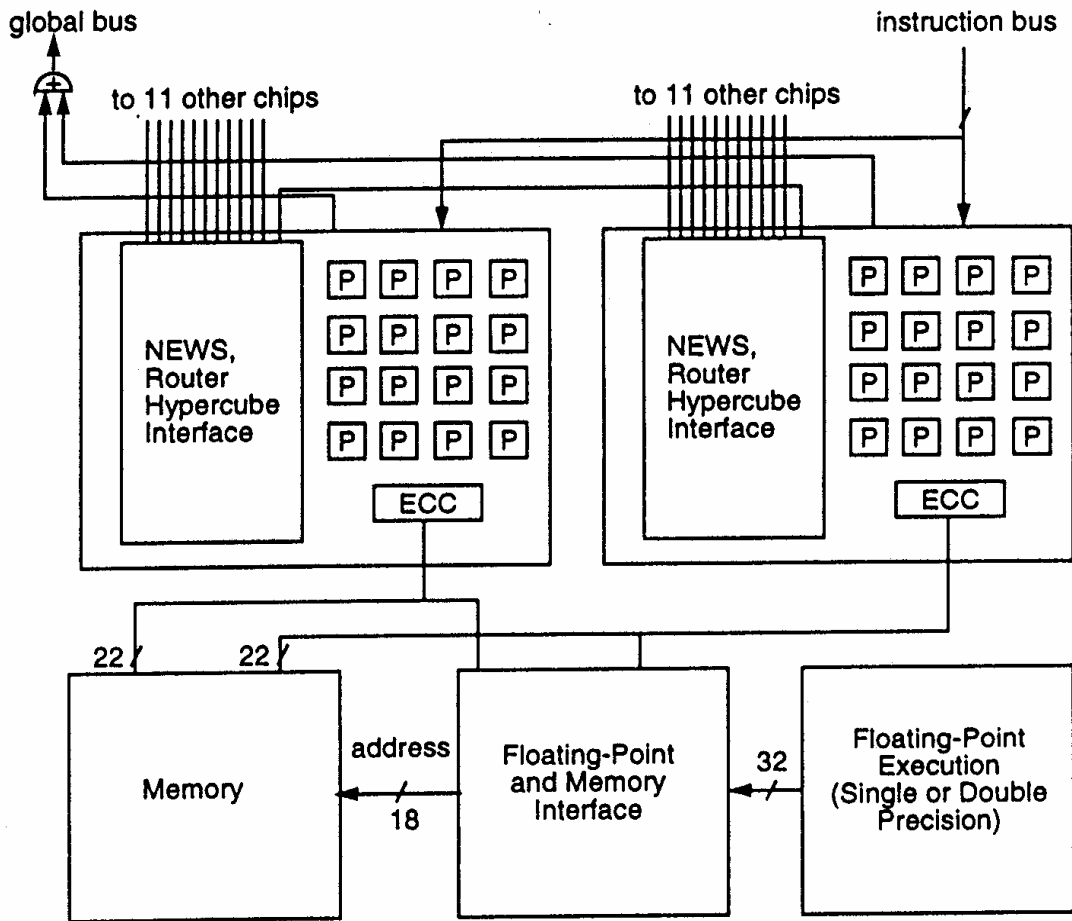
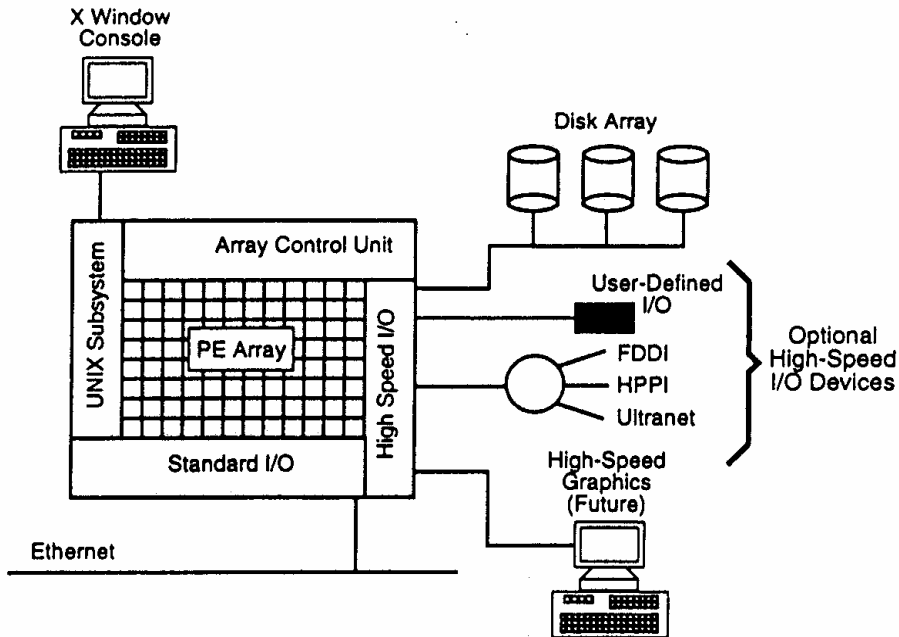


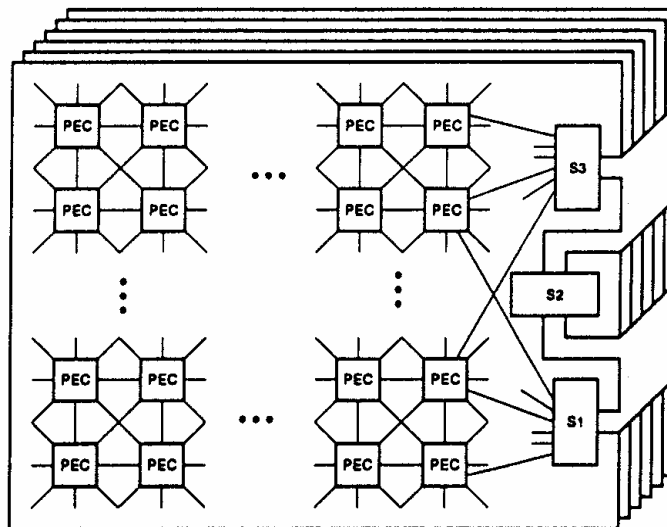
Figure 8.24 A CM-2 processing node consisting of two processor chips and some memory and floating-point chips. (Courtesy of Thinking Machines Corporation, 1990)

SIMD Computers (Cont'd)

– The MasPar MP-1 Architecture



(a) MP-1 System Block Diagram

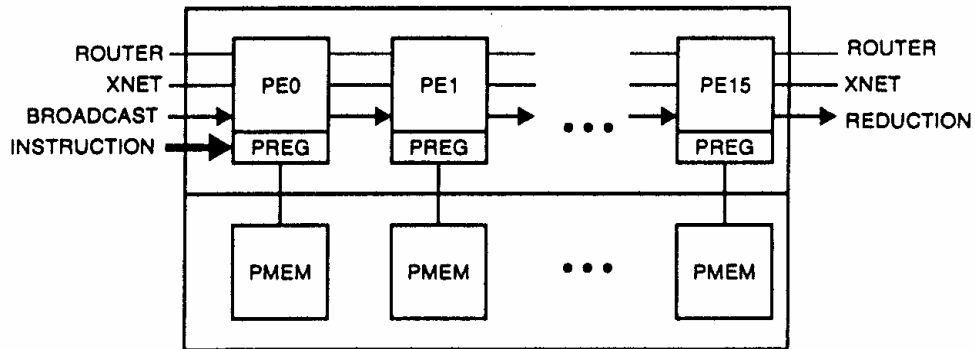


(b) Array of PE clusters

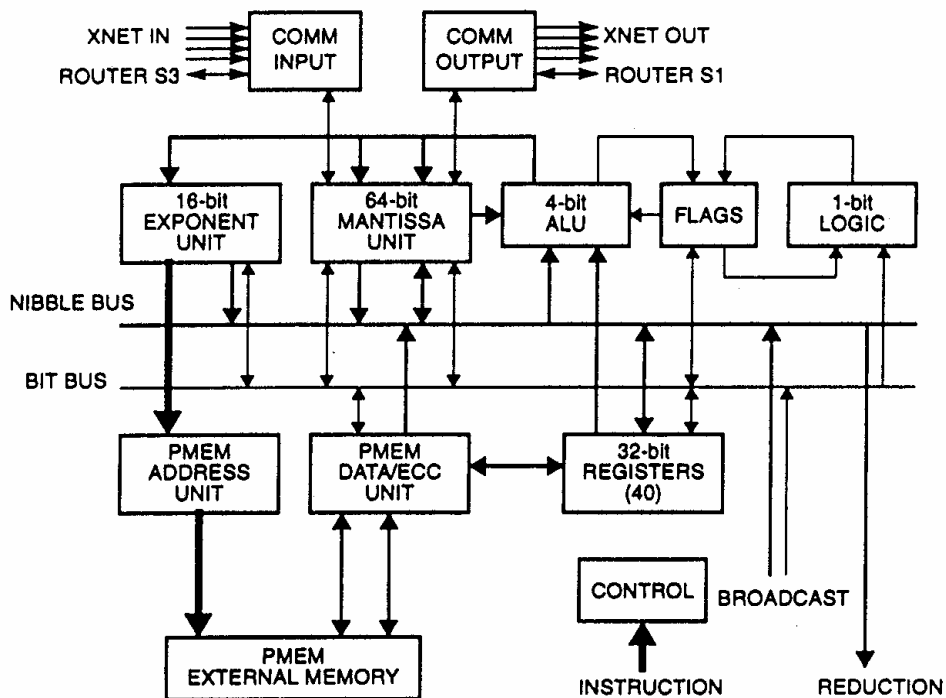
Figure 8.25 The MasPar MP-1 architecture. (Courtesy of MasPar Computer Corporation, 1990)

SIMD Computers (Cont'd)

– The MasPar MP-1 Architecture



(a) A PE cluster



(b) Processor element and memory

Figure 8.26 Processing element and memory design in the MasPar MP-1. (Courtesy of MasPar Computer Corporation, 1990)

MIMD Computers

– Bell's Taxonomy of MIMD Computers

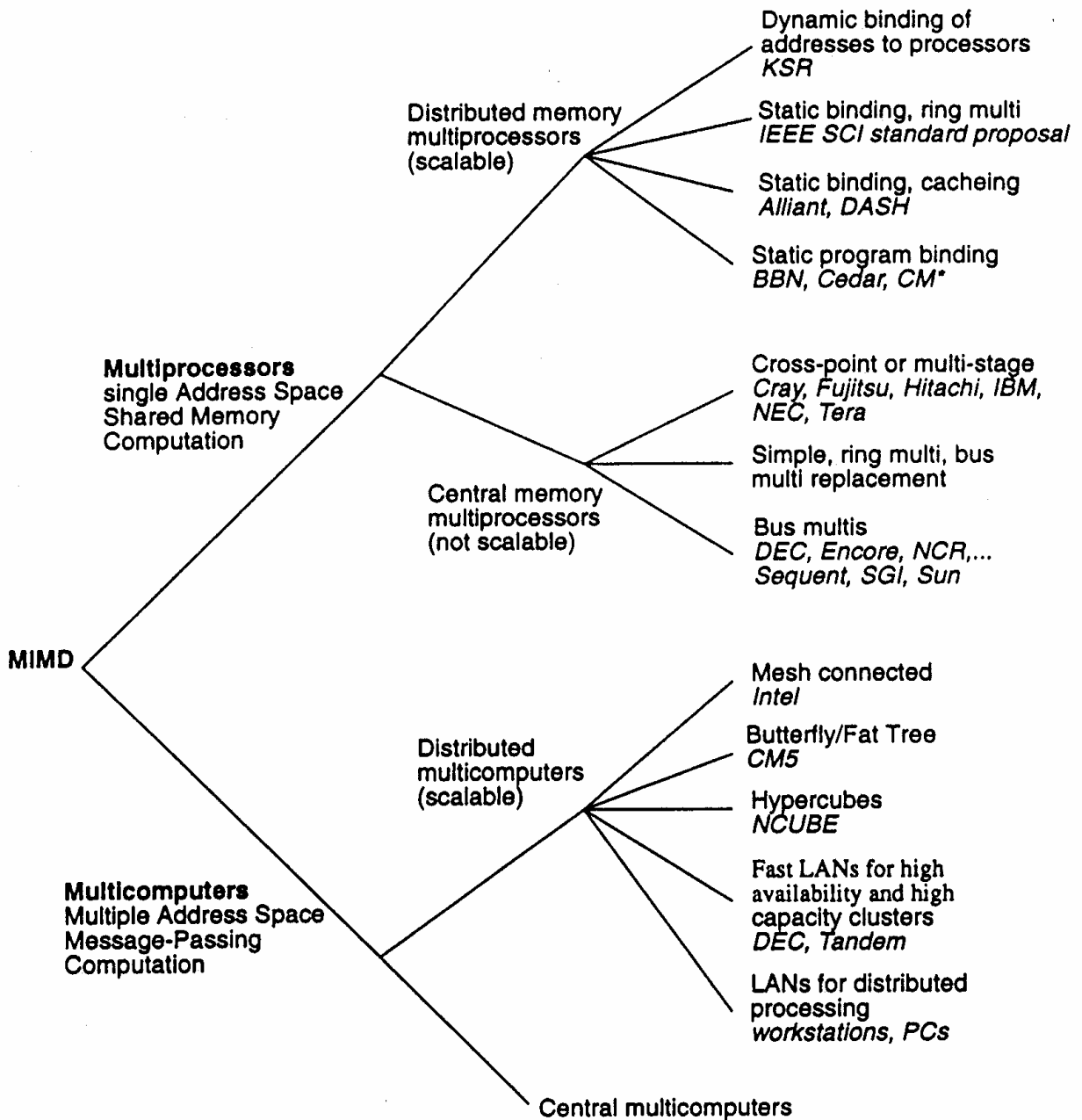
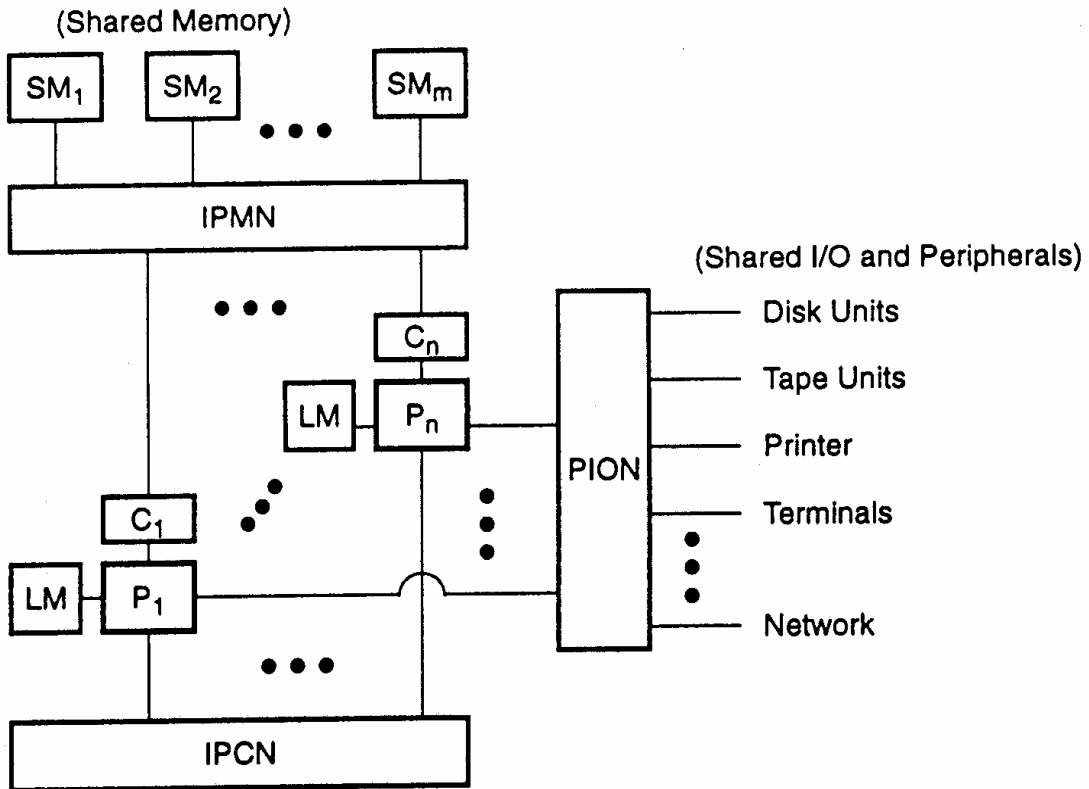


Figure 1.10 Bell's taxonomy of MIMD computers. (Courtesy of Gordon Bell; reprinted with permission from the *Communications of ACM*, August 1992)

MIMD Computers (Cont'd)

– Multiprocessors

• Generic Multiprocessor Architecture



- Legends:
- IPMN (Inter-Processor-Memory Network)
 - PION (Processor-I/O Network)
 - IPCN (Inter-Processor Communication Network)
 - P (Processor)
 - C (Cache)
 - SM (Shared Memory)
 - LM (Local Memory)

Figure 7.1 Interconnection structures in a generalized multiprocessor system with local memory, private caches, shared memory, and shared peripherals

MIMD Computers (Cont'd)

– Shared–Memory Multiprocessors

• Bus Architecture

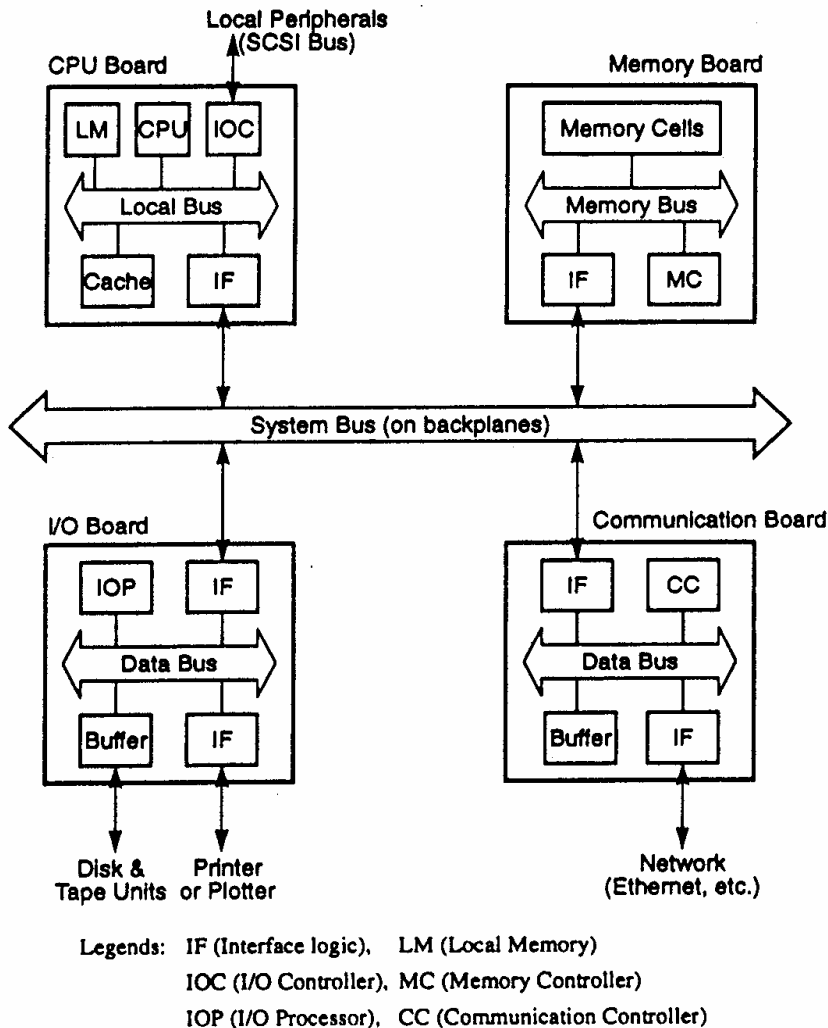


Figure 7.2 Bus systems at board level, backplane level, and I/O level.

MIMD Computers (Cont'd)

– Shared–Memory Multiprocessors

• Cross–Bar Architecture

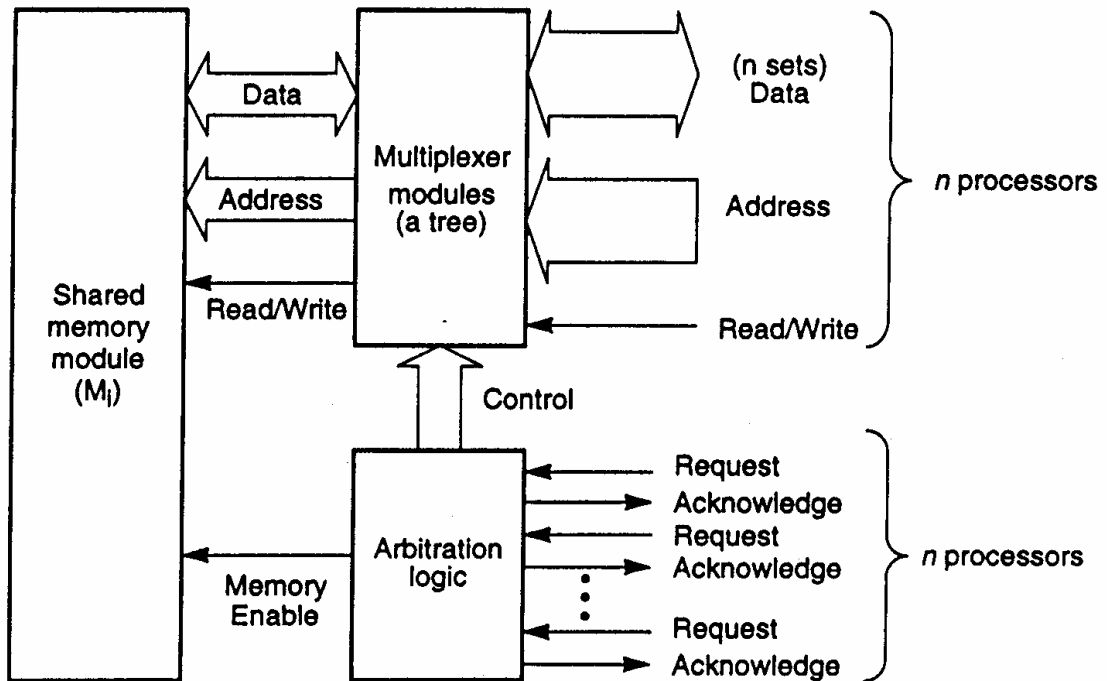
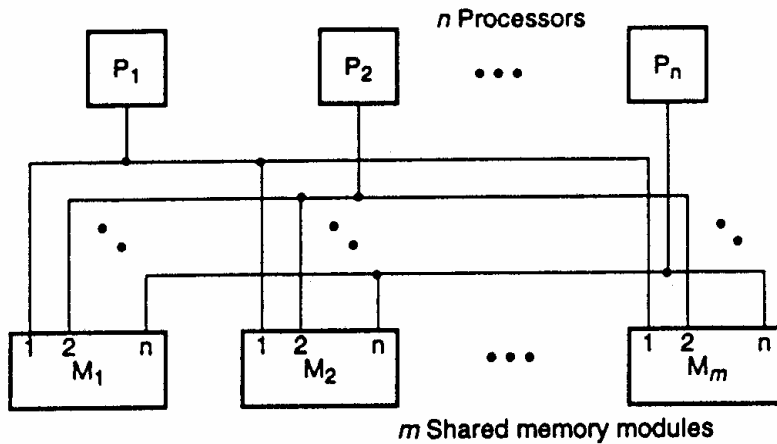


Figure 7.6 Schematic design of a crosspoint switch in a crossbar network.

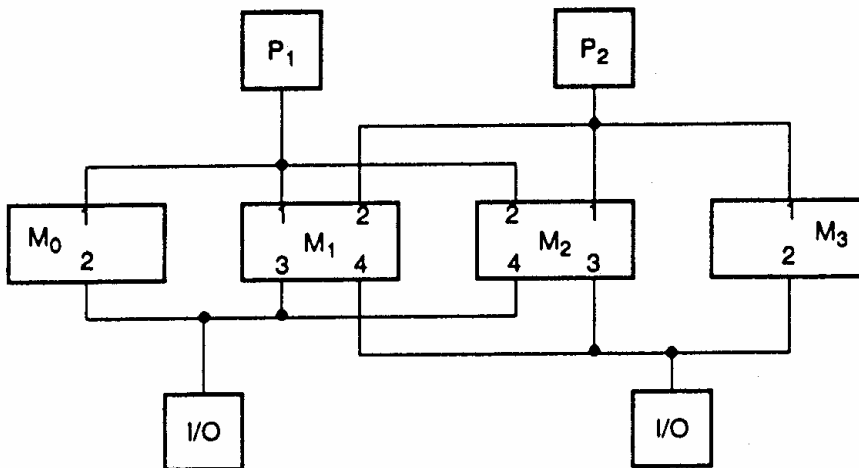
MIMD Computers (Cont'd)

– Shared–Memory Multiprocessors

• Multiport Memory Architecture



(a) n -port memory modules used



(b) Memory ports prioritized or privileged in each module by numbers

Figure 7.7 Multiport memory organizations for multiprocessor systems. (Courtesy of P. H. Enslow, *ACM Computing Surveys*, March 1977)

MIMD Computers (Cont'd)

– Distributed-Memory Multiprocessors

- KSR-1

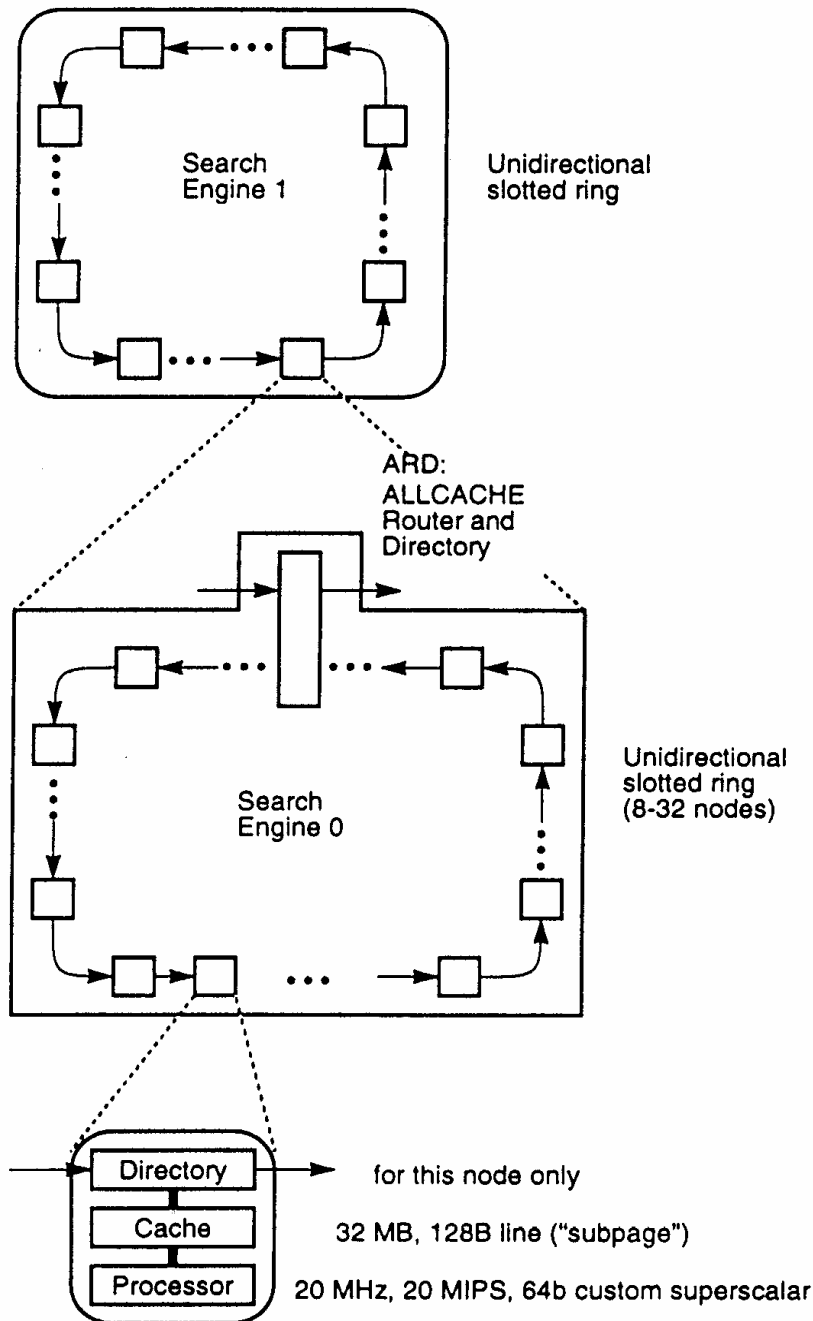


Figure 9.27 The KSR-1 architecture with a slotted ring for communication. (Courtesy of Kendall Square Research Corporation, 1991)

MIMD Computers (Cont'd)

- Distributed-Memory Multiprocessors

- KSR-1

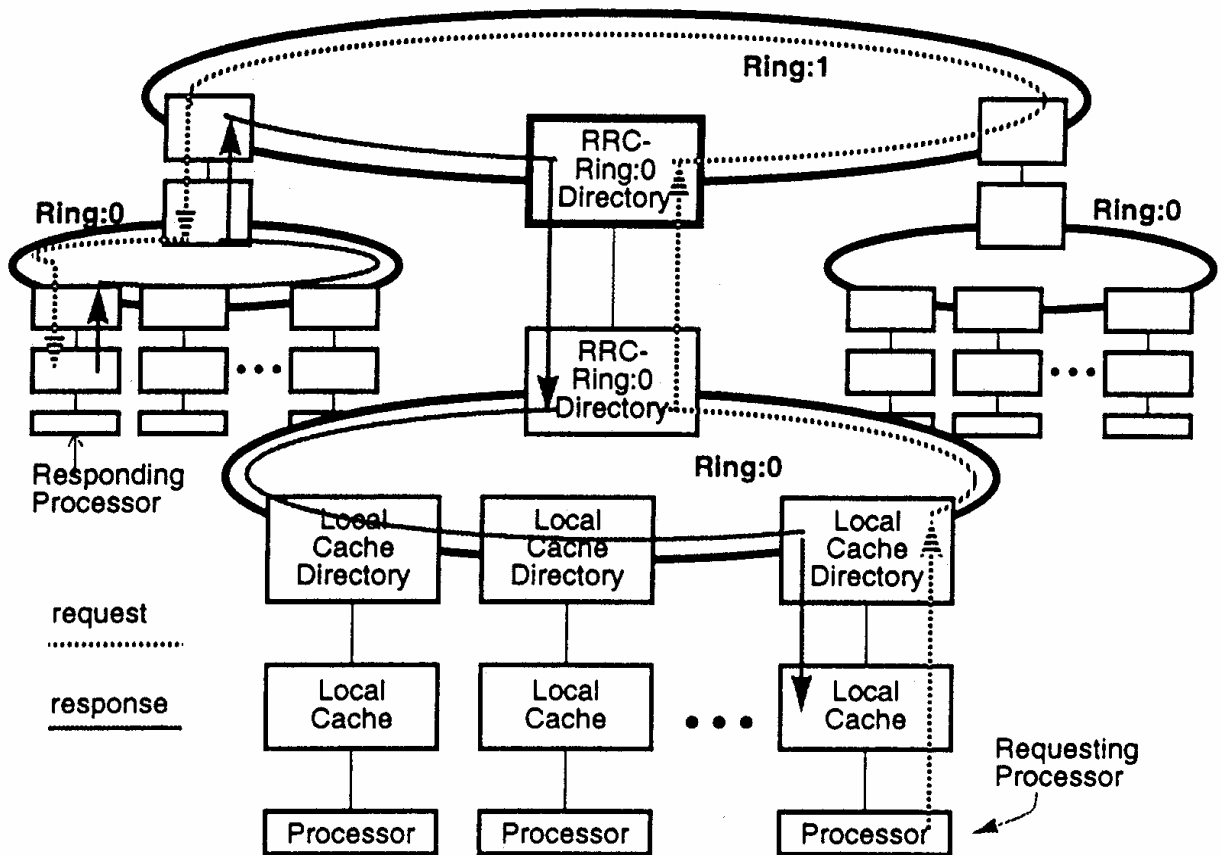


Figure 9.28 Remote cache (memory) access through two levels of communication rings in the KSR-1. (Courtesy of Kendall Square Research Corporation, 1991)

MIMD Computers (Cont'd)

– Representative Multiprocessors Systems

Table 1.3 Representative Multiprocessor Systems

Company and Model	Hardware and Architecture	Software and Applications	Remarks
Sequent Symmetry S-81	Bus-connected with 30 i386 processors, IPC via SLIC bus; Weitek floating-point accelerator.	DYNIX/OS, KAP/Sequent preprocessor, transaction multiprocessing.	i486-based multiprocessor available 1991. i586-based systems to appear.
IBM ES/9000 Model 900/VF	Model 900/VF has 6 ES/9000 processors with vector facilities, crossbar connected to I/O channels and shared memory.	OS support: MVS, VM KMS, AIX/370, parallel Fortran, VSF V2.5 compiler.	Fiber optic channels, integrated cryptographic architecture.
BBN TC-2000	512 M88100 processors with local memory connected by a Butterfly switch, a NUMA machine.	Ported Mach/OS with multiclustering, use parallel Fortran, time-critical applications.	Shooting for higher performance using faster processors in future models.

MIMD Computers (Cont'd)

– Multicomputers

- Generic Model of a Multicomputer

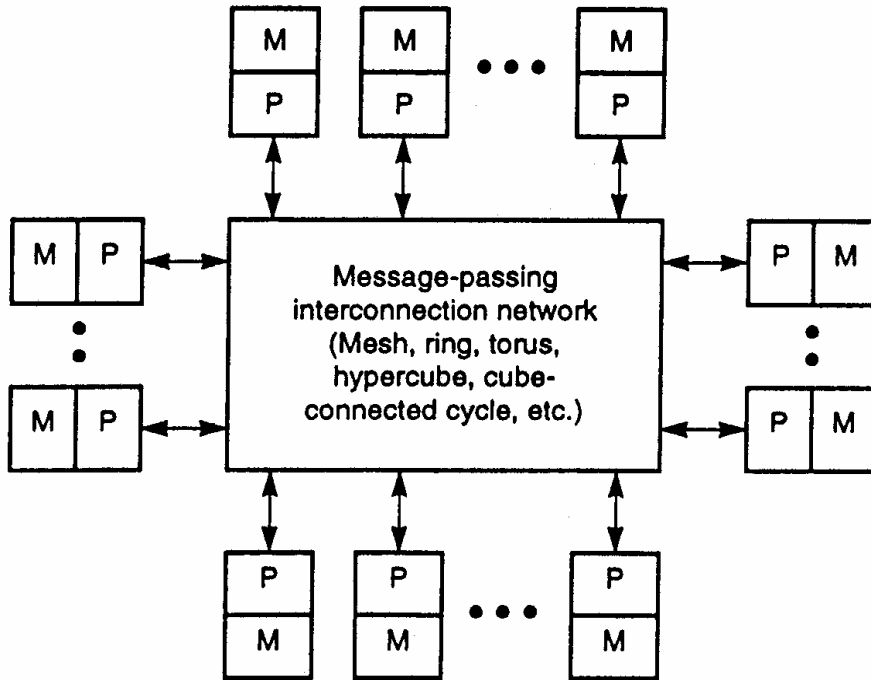


Figure 1.9 Generic model of a message-passing multicomputer.

MIMD Computers (Cont'd)

– Multicomputers

• Representative Multicomputer Systems

Table 1.4 Representative Multicomputer Systems

System Features	Intel Paragon XP/S	nCUBE/2 6480	Parsys Ltd. SuperNode1000
Node Types and Memory	50 MHz i860 XP computing nodes with 16–128 Mbytes per node, special I/O service nodes.	Each node contains a CISC 64-bit CPU, with FPU, 14 DMA ports, with 1–64 Mbytes/node.	EC-funded Esprit supernode built with multiple T-800 Transputers per node.
Network and I/O	2-D mesh with SCSI, HIPPI, VME, Ethernet, and custom I/O.	13-dimensional hypercube of 8192 nodes, 512-Gbyte memory, 64 I/O boards.	Reconfigurable interconnect, expandable to have 1024 processors.
OS and Software task parallelism Support	OSF conformance with 4.3 BSD, visualization and programming support.	Vertex/OS or UNIX supporting message passing using wormhole routing.	IDRIS/OS is UNIX-compatible, supported.
Application Drivers	General sparse matrix methods, parallel data manipulation, strategic computing.	Scientific number crunching with scalar nodes, database processing.	Scientific and academic applications.
Performance Remarks	5–300 Gflops peak 64-bit results, 2.8–160 GIPS peak integer performance.	27 Gflops peak, 36 Gbytes/s I/O, challenge to build even larger machine.	200 MIPS to 13 GIPS peak, largest supernode in use contains 256 Transputers.

MIMD Computers (Cont'd)

– Multicomputers

• Paragon Architecture

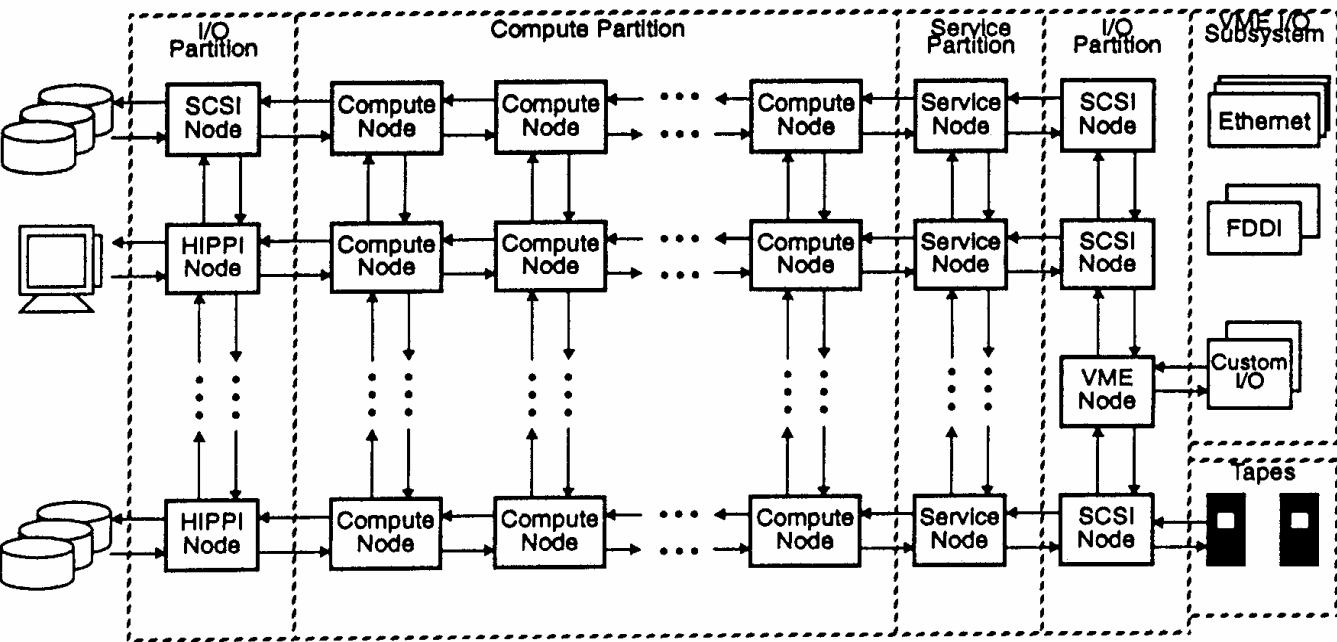


Figure 7.24 The Intel Paragon system architecture. (Courtesy of Intel Supercomputer Systems Division, 1991)

MIMD Computers (Cont'd)

– Multicomputers

• Paragon Architecture

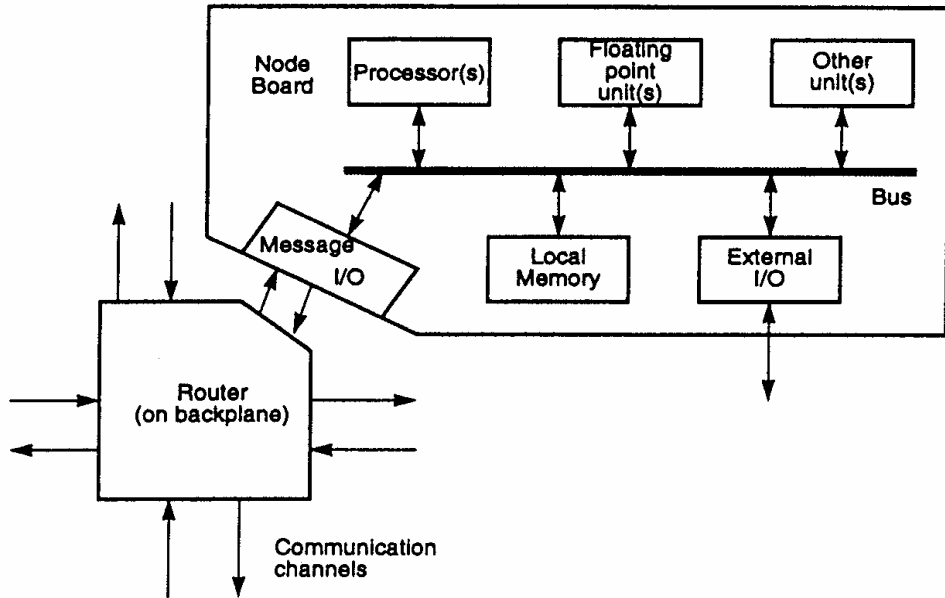


Figure 7.25 Node architecture of the Paragon multicomputer.

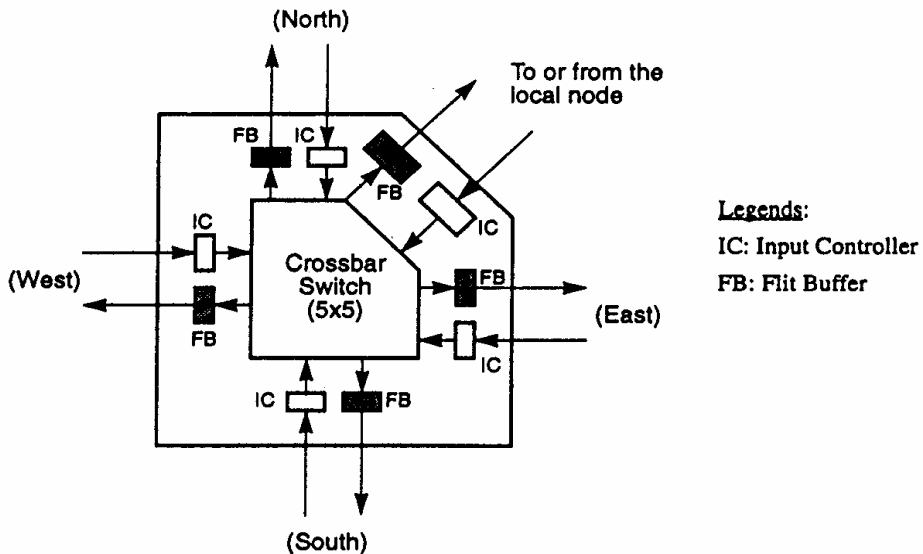


Figure 7.26 The structure of a mesh-connected router with four pairs of I/O channels connected to neighboring routers.

MIMD Computers (Cont'd)

– Multicomputers

- CM5 Architecture

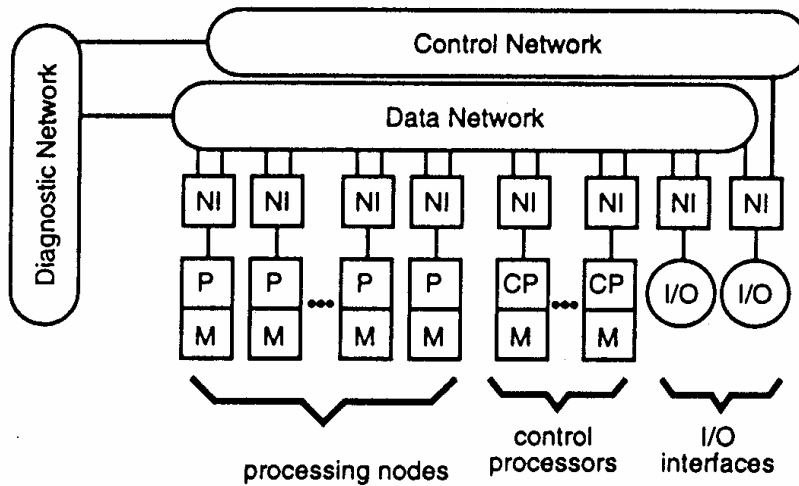


Figure 8.27 The network architecture of the Connection Machine CM-5. (Courtesy of Leiserson et al., Thinking Machines Corporation, 1992)

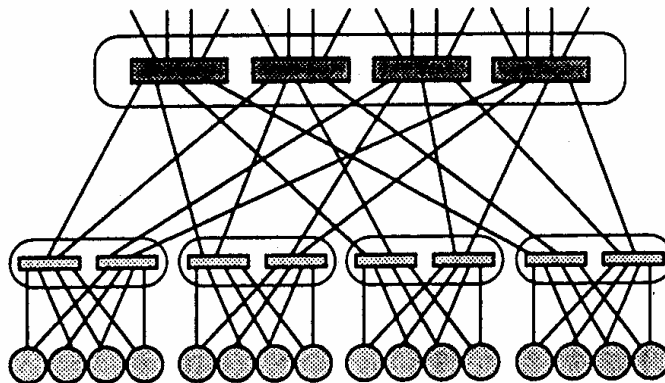


Figure 8.29 CM-5 data network implemented with a 4-ary fat tree (Courtesy of Leiserson et al., Thinking Machines Corporation, 1992)

MIMD Computers (Cont'd)

– Multicomputers

- CM5 Architecture

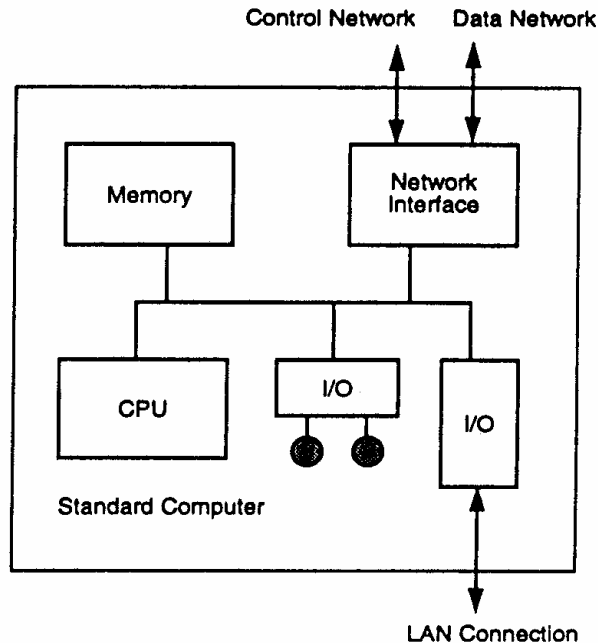


Figure 8.30 The control processor in the CM-5. (Courtesy of Thinking Machines Corporation 1992)

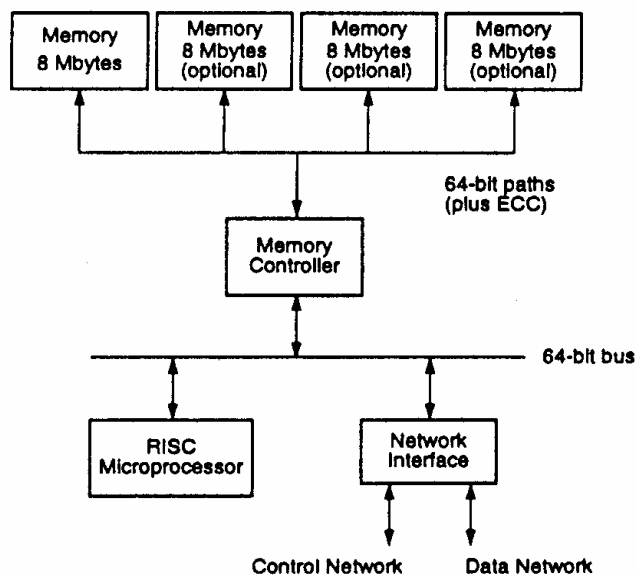
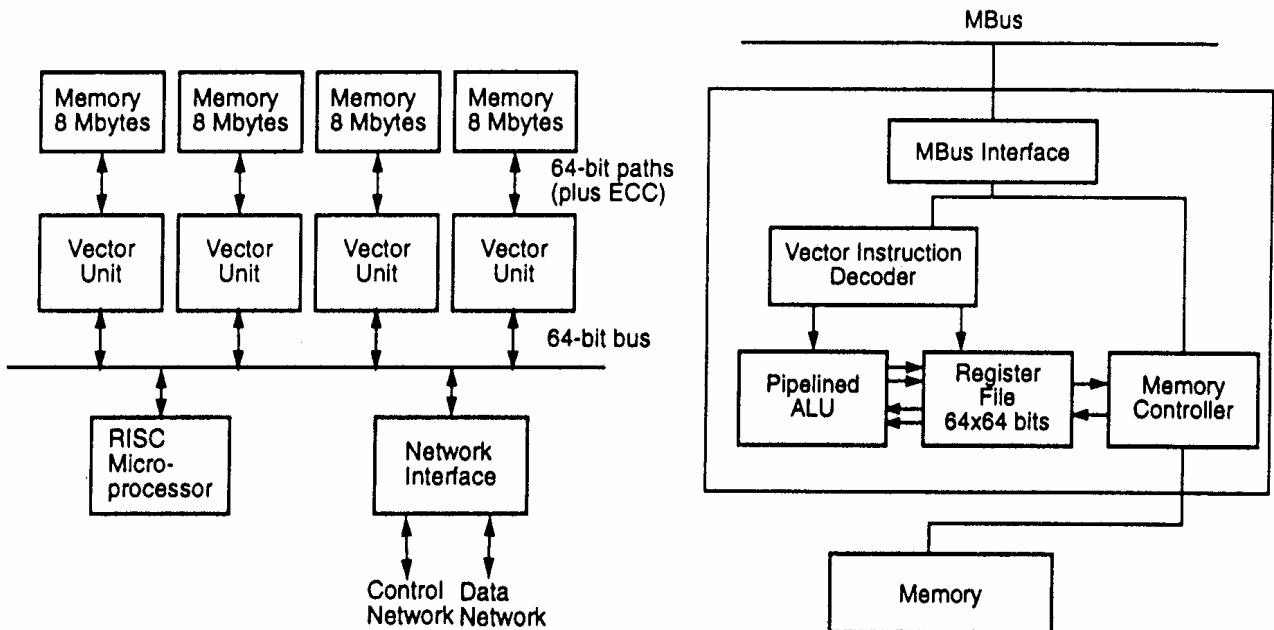


Figure 8.31 The processing node in the CM-5. (Courtesy of Thinking Machines Corporation, 1992)

MIMD Computers (Cont'd)

– Multicomputers

• CM5 Architecture



(a) Processing node with vector units

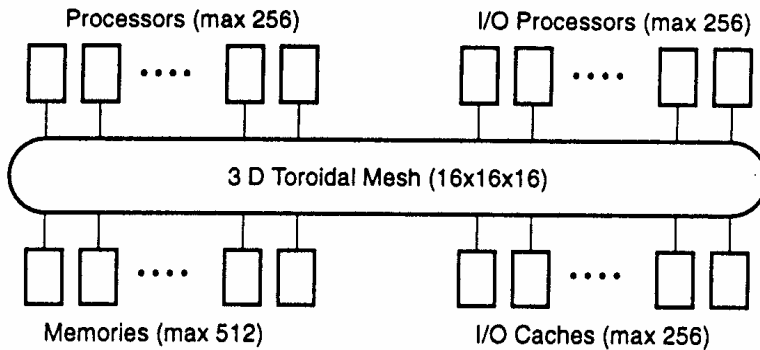
(b) Vector unit functional architecture

Figure 8.32 The processing node with vector units in the CM-5. (Courtesy of Thinking Machines Corporation, 1992)

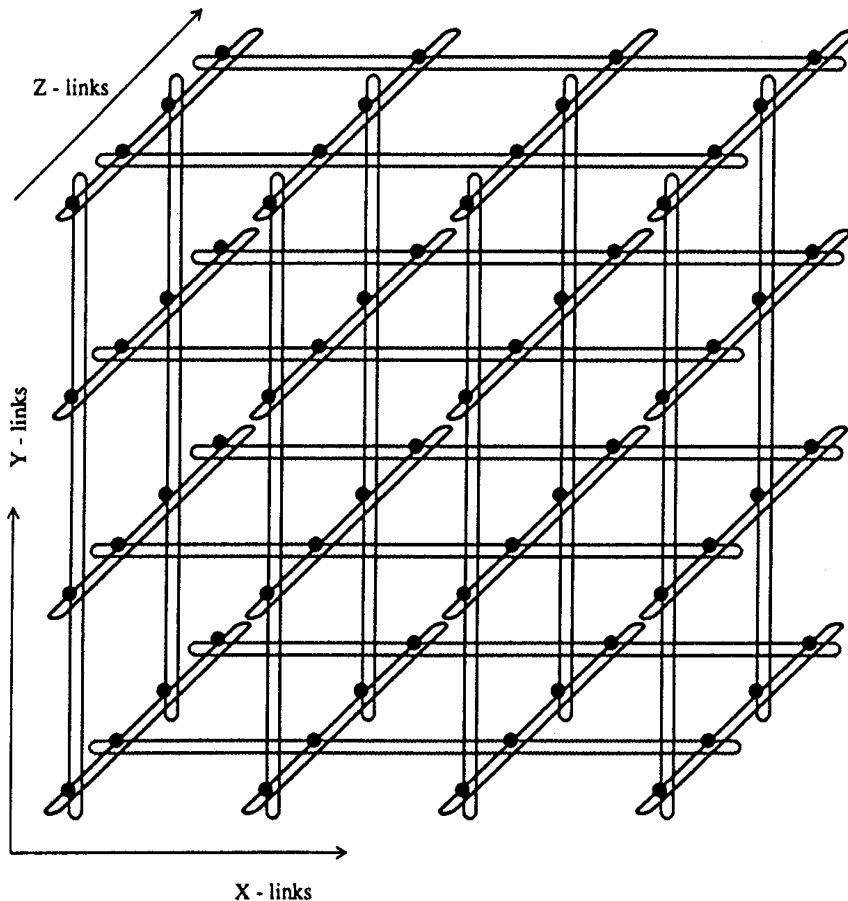
MPP Computers (Cont'd)

– Architectures for MPP

- Scalable Architecture



(a) The Tera computer system



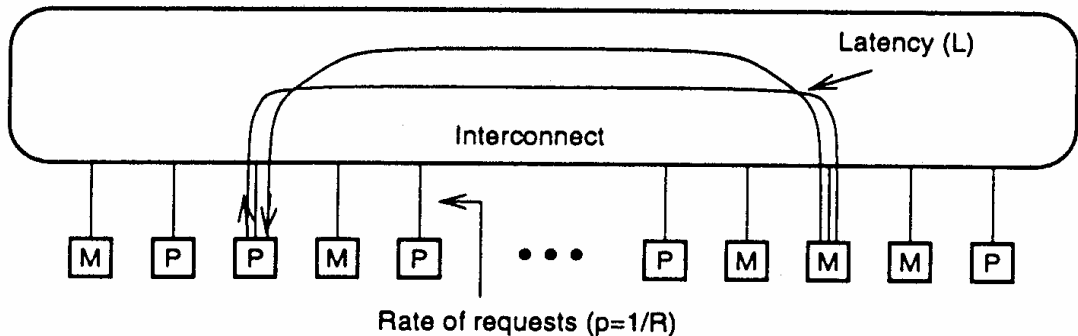
(b) A sparse $4 \times 4 \times 4$ torus with X-links and Y-links missing on alternate Z-layers, respectively

Figure 9.29 The Tera multiprocessor and its three-dimensional sparse torus architecture shown with a $4 \times 4 \times 4$ configuration. (Courtesy of Tera Computer Company. 1992)

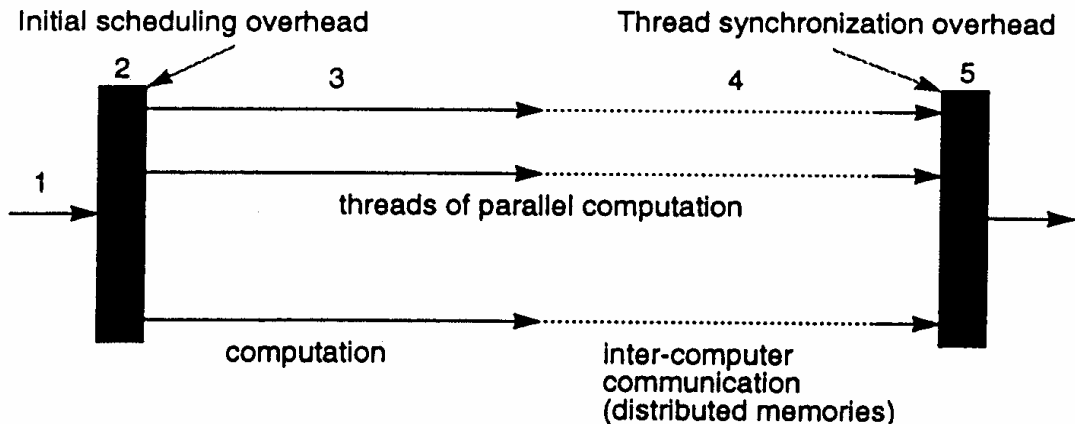
MPP Computers (Cont'd)

– Architectures for MPP

- Multithreaded Architecture



(a) The architecture environment. (Courtesy of Rafael Saavedra, 1992)



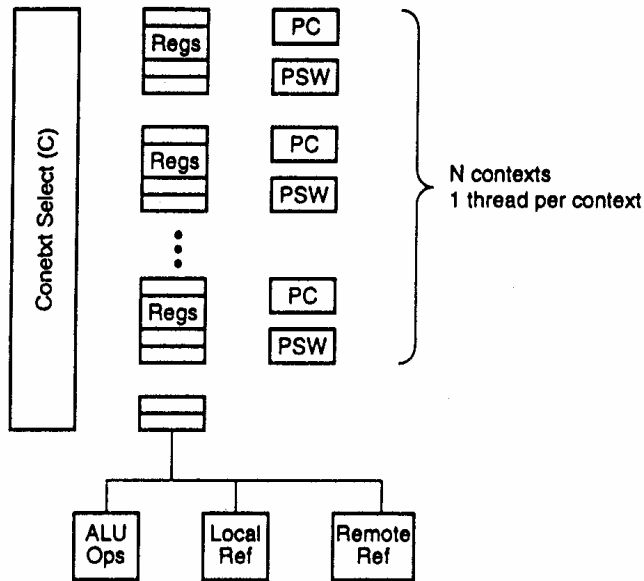
(b) Multithreaded computation model. (Courtesy of Gordon Bell, *Commun. ACM*, August 1992)

Figure 9.11 Multithreaded architecture and its computation model for a massively parallel processing system.

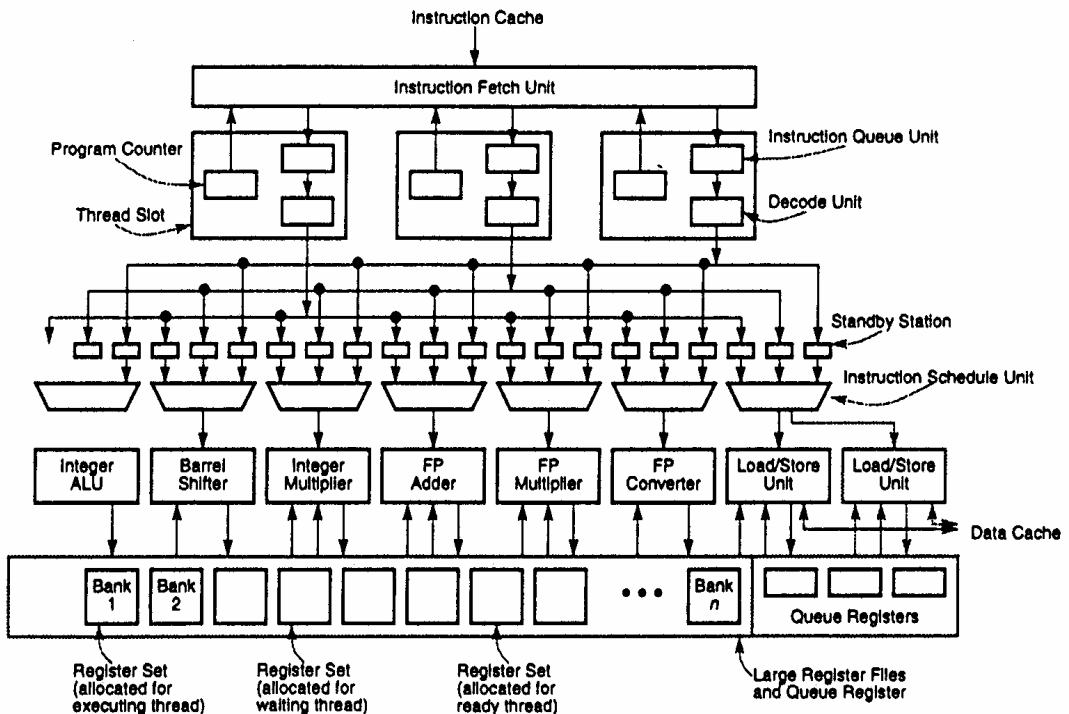
MPP Computers (Cont'd)

– Architectures for MPP

• Multithreaded Architecture



(a) Multithreaded model. (Courtesy of Rafael Saavedra, 1992)



(b) A three-thread processor example. (Courtesy of H. Hirata et al., *Proc. 19th Int. Symp. Comput. Archit.*, Australia, May 1992)

Figure 9.14 Multiple-context processor model and an example design.

MPP Computers (Cont'd)

– Architectures for MPP

• Dataflow Architecture

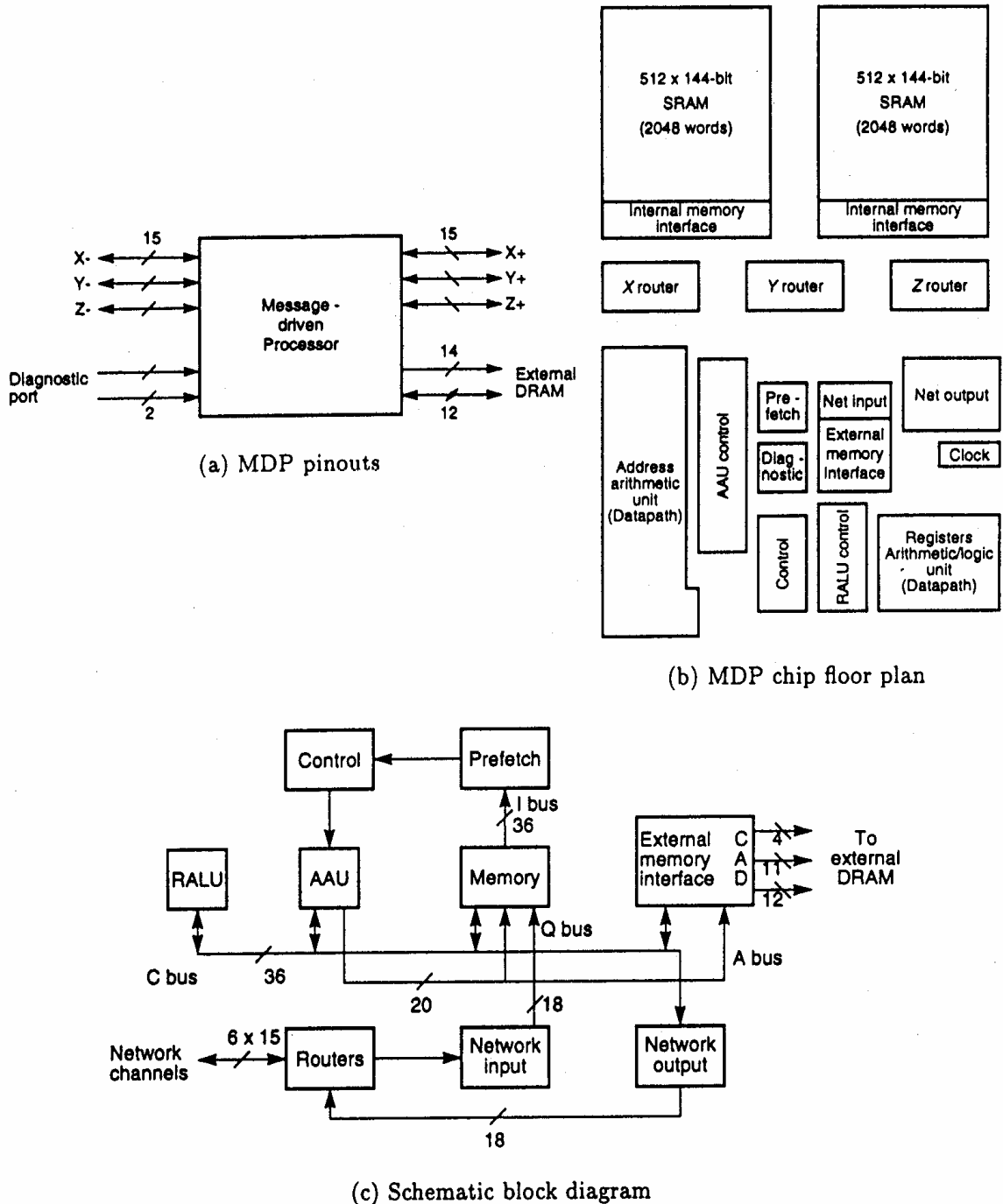


Figure 9.19 The message-driven processor (MDP) architecture. (Courtesy of W. Dally et al.; reprinted with permission from *IEEE Micro*, April 1992)